

Computer Organisation

Topic 9

Interfacing and Communication

Outline

- External Devices
- I/O Modules
- I/O Operations
 - Programmed I/O, Interrupt-Driven I/O and Direct Memory Access (DMA)
- I/O Channels and Processors
- Interfaces
 - Parallel and serial interfaces
- Introduction to Networks
- RAID architectures

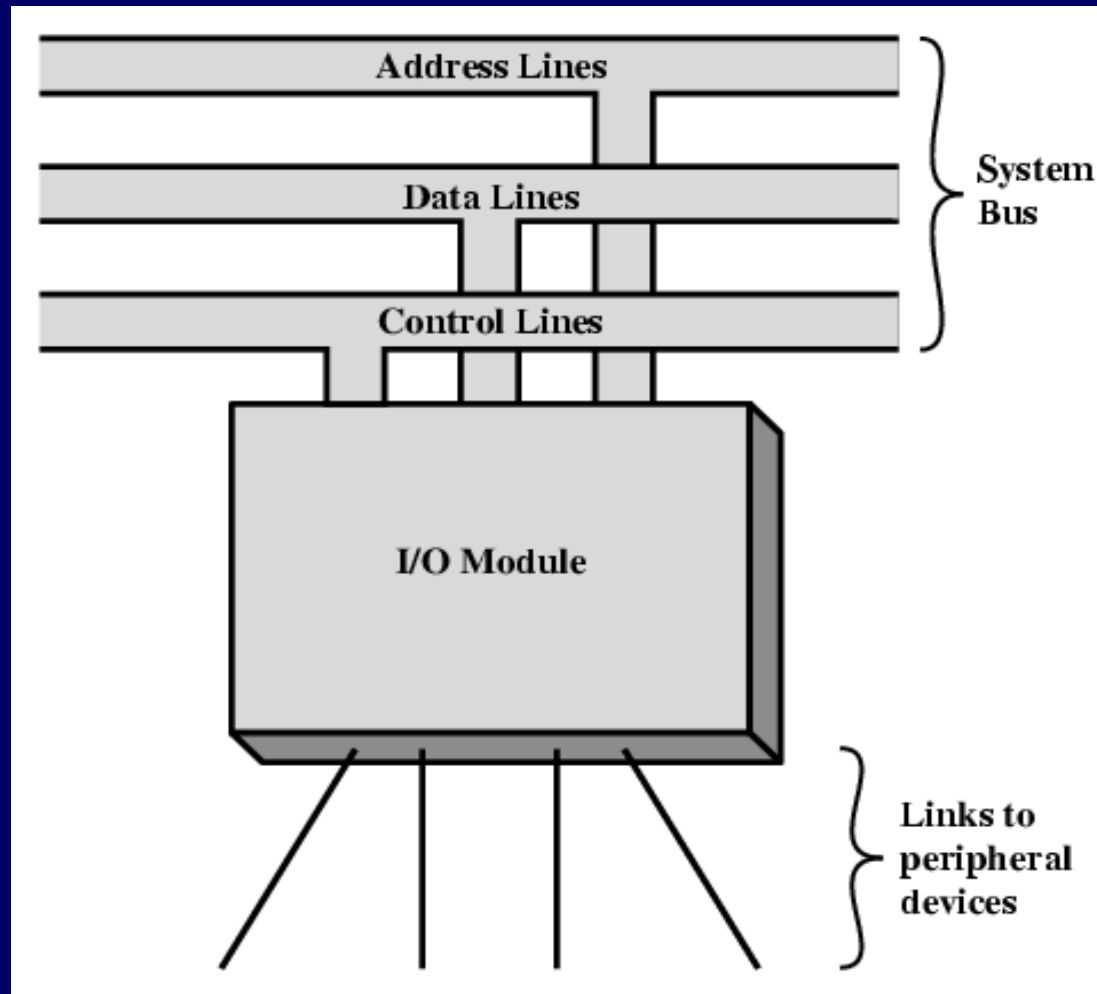
Input/Output Problems

- Why peripherals are not directly connected to system bus?
 - Wide variety of peripherals with various methods of operation
 - Delivering different amounts of data
 - At different speeds
 - In different formats and word lengths
- All slower than CPU and RAM
- Need I/O modules

Input/Output Module

- Interface to CPU and Memory via the system bus
- Interface to one or more peripherals by tailored data links
- Is designed
 - to provide a systematic means of controlling interaction between the devices and CPU/memory
 - To manage the I/O activity

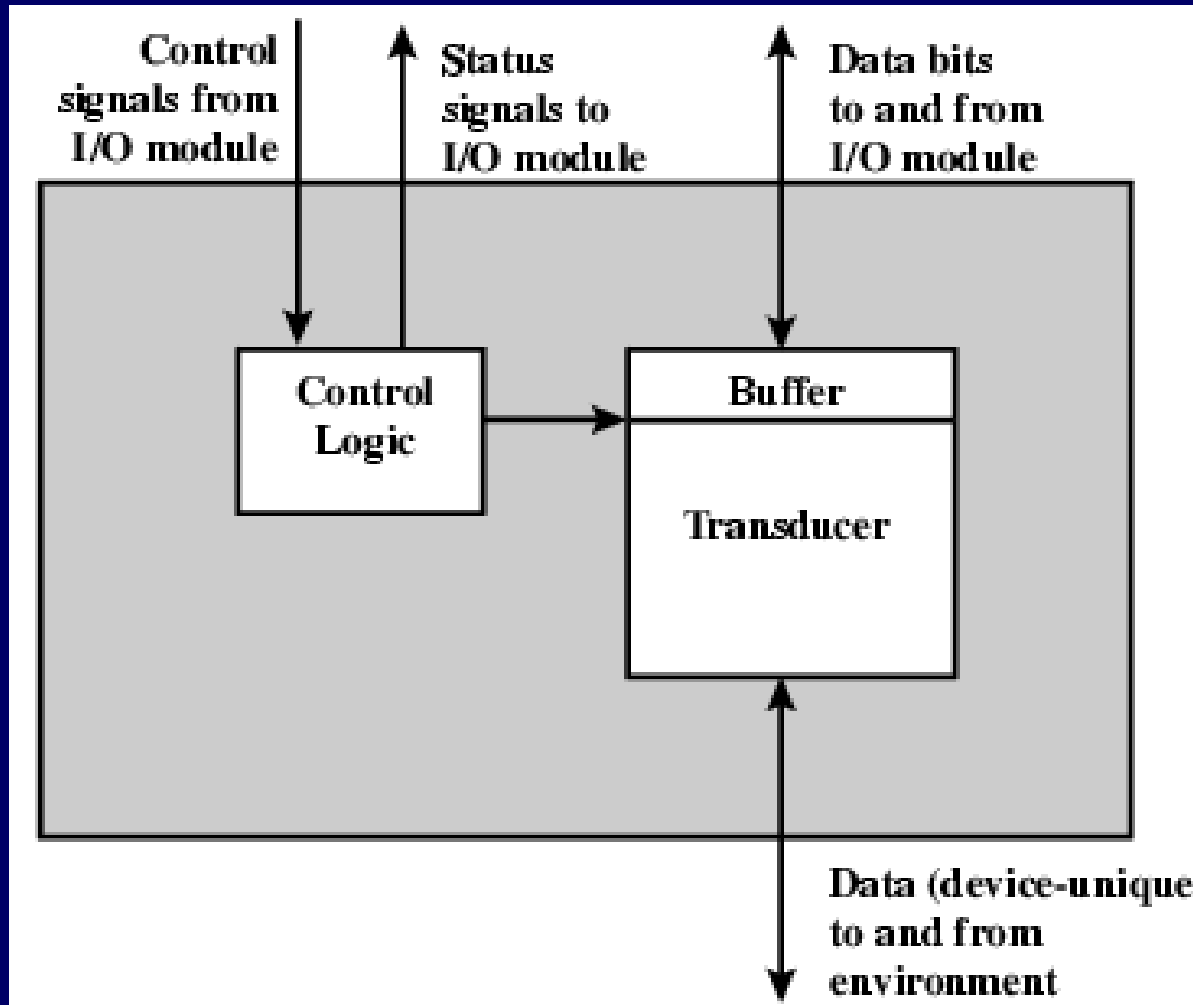
Generic Model of I/O Module



External Devices

- External device attaches to the computer by a link to the I/O module
 - To exchange control, status, and data
- Categories of external devices
 - Human readable
 - Screen, printer, keyboard, etc.
 - Machine readable
 - Monitoring and control
 - Magnetic disks, etc.
 - Communication
 - Modem, Network Interface Card (NIC), etc.

External Device Block Diagram



Control logic

Controls the device's operation in response to direction from I/O module

Transducer

Converts data from electrical to other forms of energy during output, and from other forms to electrical during input

External Device Interface

- Control, data, and status
- Control signals
 - Determine function the device will perform
 - E.g. send data to I/O module (INPUT/READ), accept data from I/O module (OUTPUT/WRITE), report status, etc.
- Data
 - A set of bits to be send/received from I/O module
- Status
 - Indicate state of device
 - E.g. READY/NOT-READY (for data transfer), etc.

Example with Keyboard/Monitor

- Common text code is the International Reference Alphabet (IRA)
- For letter "K", IRA code is 1001011
- For keyboard input
 - User presses "K" key
 - Generate electronic signal
 - Interpreted by transducer, translated into 1001011
 - 1001011 transmitted to I/O module in the computer
- For monitor output
 - From I/O module, 1001011 sent to monitor
 - Transducer at monitor interprets 1001011
 - Monitor sends electronic signals to display letter "K"

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I/O Module Function

- Control & Timing
- CPU Communication
- Device Communication
- Data Buffering
- Error Detection

Control and Timing

- CPU checks I/O module device status
- I/O module returns the device status
- If the device ready, CPU requests data transfer
- I/O module gets data (8 or 16 bits) from device
- I/O module transfers data to CPU

Processor Communication

- Command decoding
 - I/O module accepts commands from the processor
 - E.g. for disk drive: READ SECTOR, WRITE SECTOR, SEEK track number, scan record ID
- Data
 - Data are exchanged between the CPU and I/O module over the data bus
- Status reporting
 - Need to know status of I/O module because devices are slow
 - The devices is ready or not: READY or BUSY
- Address recognition
 - I/O module must recognize one unique address for each device

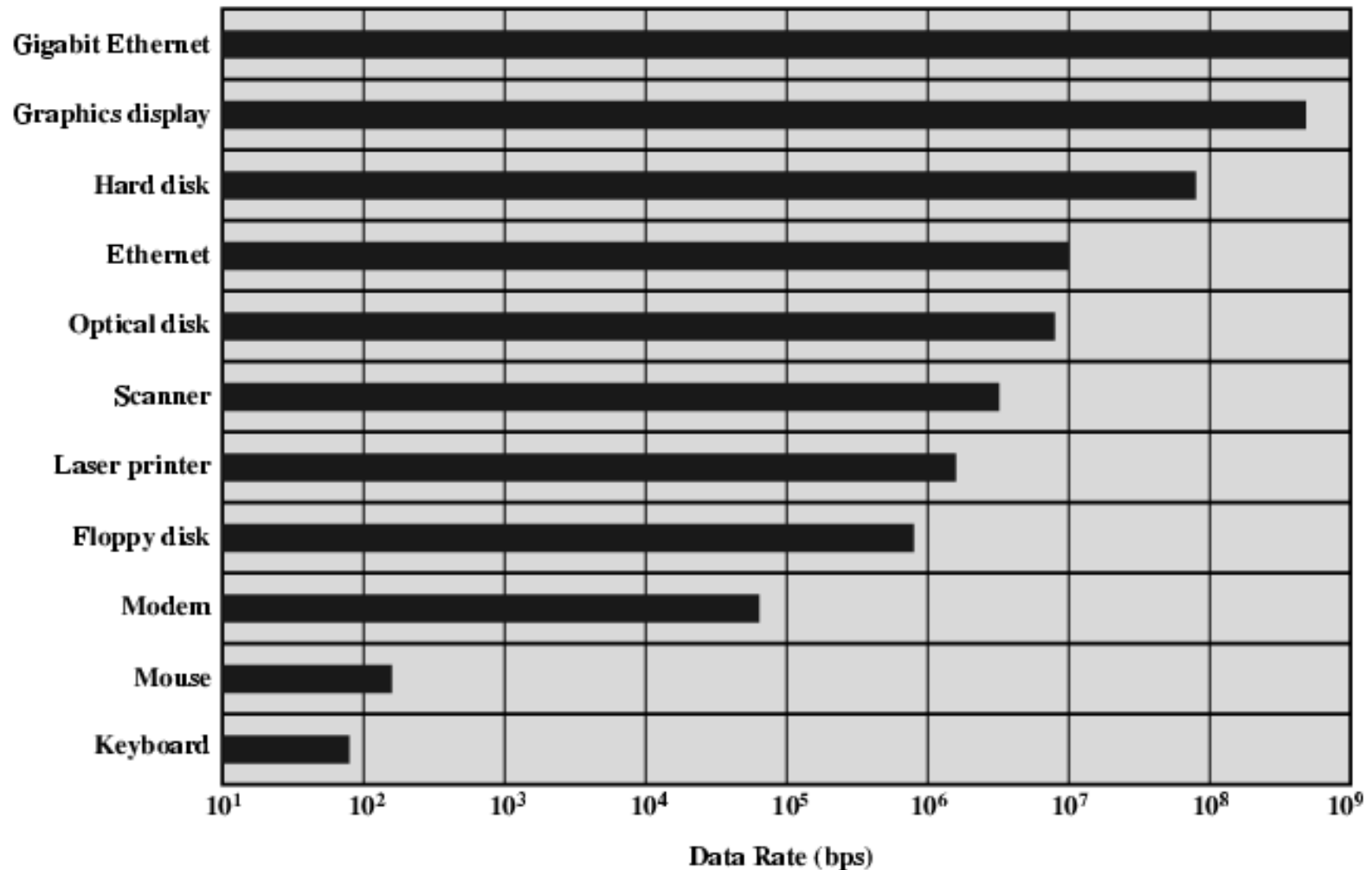
Device Communication

- Related to device interface
- Communication involves
 - Command/control
 - Status
 - Data

Data Buffering

- The transfer rate into and out of main memory/CPU is quite high
- The data are buffered in the I/O module and then sent to the device at its data rate
- I/O module must be able to operate at both devices and memory speeds

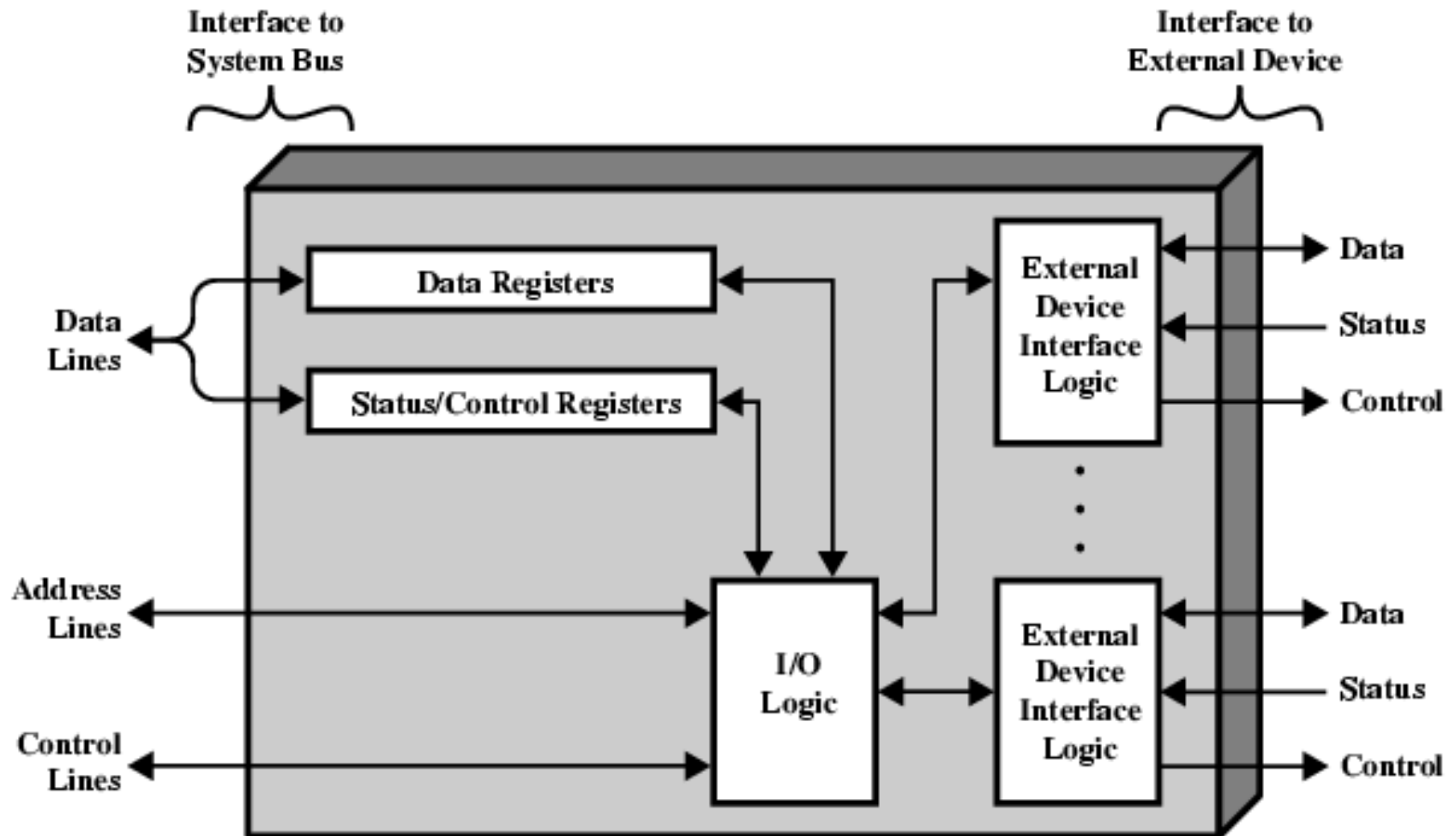
Typical I/O Data Rates



Error Detection

- Report errors to the CPU
 - Mechanical/electrical malfunction
 - Paper jam, bad disk track
 - Changes in bit pattern during transmission
 - Use error detecting code: parity bit
 - E.g. IRA characters occupies 7 bits of a byte. 8th bit is the parity bit (odd/even)

I/O Module Diagram



I/O Module Decisions

- Hide the details of timing, formats or reveal device properties to CPU
- Support multiple or single device
- If I/O module controls most of device functions
 - I/O channel or I/O processor
 - In mainframe computers
- If I/O module is primitive
 - Leave much work for CPU
 - I/O controller or device controller

I/O Module

I/O Channel or
I/O Processor

I/O Controller or
Device Controller

Input Output Operations/Techniques

- Programmed
- Interrupt driven
- Direct Memory Access (DMA)

	No Interrupt	Use Interrupt
I/O-to-memory transfer through processor	Programmed I/O	Interrupt-driven I/O
Direct I/O-to-memory transfer		Direct Memory Access (DMA)

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Programmed I/O

- When I/O-related instruction encountered in program, processor issues command to appropriate I/O module
- CPU has direct control over I/O operation
 - Sensing device status
 - Sending Read/write commands
 - Transferring data
- CPU waits for I/O module to complete operation
 - I/O module does not interrupt processor to alert
- Wastes CPU time

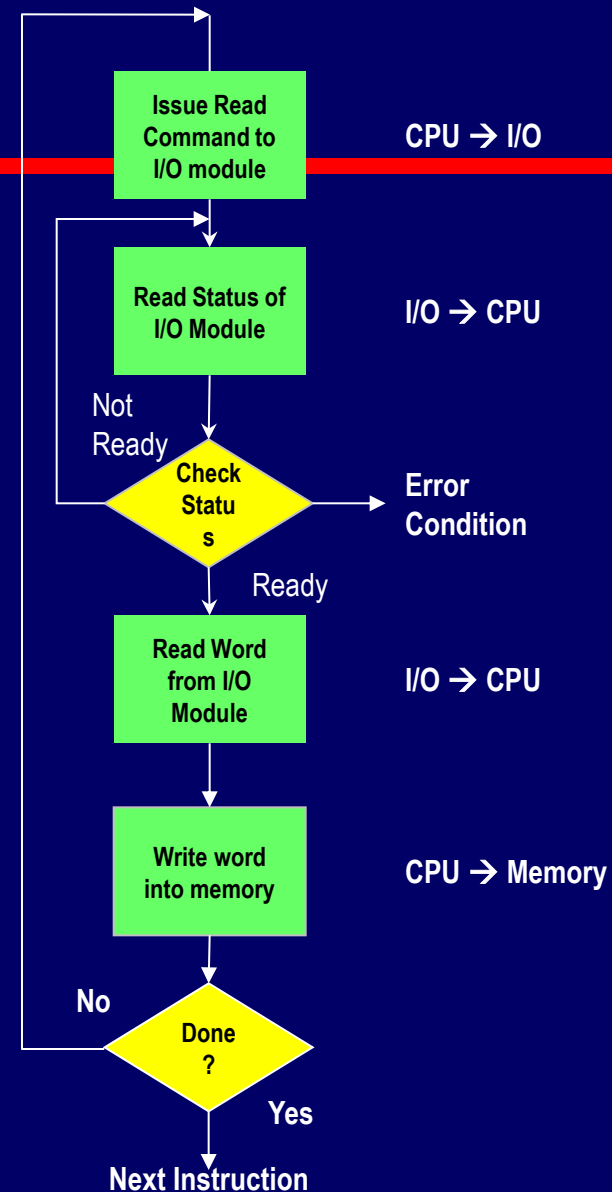
Programmed I/O – what happens

- CPU issue a command/requests I/O operation
- I/O module performs the requested operation
- I/O module sets status bits in the I/O status register
- CPU checks status bits periodically
- I/O module does not inform CPU directly
- I/O module does not interrupt CPU
- CPU may wait or come back later

I/O Commands

- To execute I/O-related instruction, CPU issues
 - address (of I/O module/device)
 - an I/O command
- CPU issues address
 - Identifies module (& device if >1 per module)
- CPU issues command
 - Control - telling I/O module what to do
 - e.g. spin up disk
 - Test - check status
 - e.g. power? error?
 - Read
 - I/O module obtains data from device and places it in an internal buffer
 - Write
 - I/O module takes data from the data bus and transmits to the device

Programmed I/O - for Input



Addressing I/O Devices

- Under programmed I/O data transfer is very like memory access (CPU viewpoint)
- Each device given unique identifier
- CPU commands contain identifier (address)
- When processor, memory, and I/O share a common bus, 2 modes of addressing:
 - Memory-mapped I/O
 - Isolated I/O

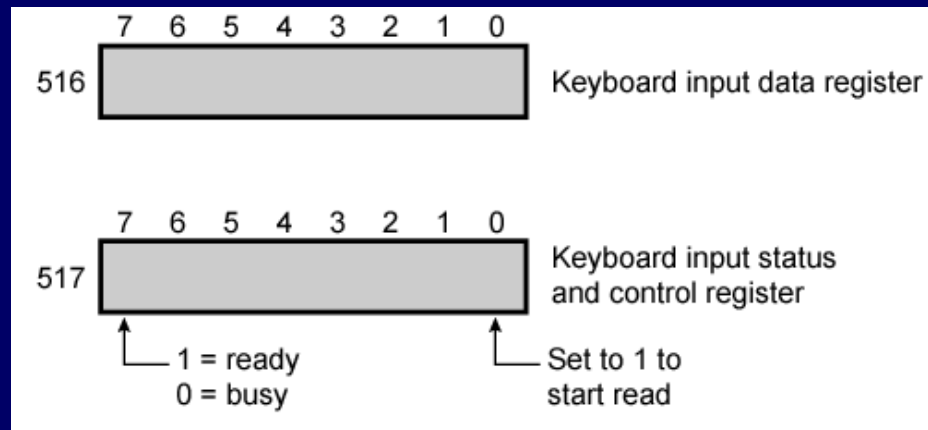
I/O Mapping

- Memory mapped I/O
 - Single read line and single write line on bus
 - Devices and memory share an address space (single address space)
 - I/O looks just like memory read/write
 - No special commands for I/O
 - Large selection of memory access commands available
 - But more memory address space used up
 - E.g. with 10 address lines, $2^{10} = 1024$ memory locations and I/O addresses can be supported

I/O Mapping

- Isolated I/O
 - Bus equipped with memory read/write lines and I/O command lines
 - Separate address spaces for memory and I/O
 - Need I/O or memory select lines
 - Special commands for I/O
 - Limited set
 - E.g. with 10 address lines, $2^{10} = 1024$ memory locations and $2^{10} = 1024$ I/O addresses can be supported

I/O Mapping



ADDRESS	INSTRUCTION	OPERAND	COMMENT
200	Load AC	"1"	Load accumulator
	Store AC	517	Initiate keyboard read
202	Load AC	517	Get status byte
	Branch if Sign = 0	202	Loop until ready
	Load AC	516	Load data byte

(a) Memory-mapped I/O

ADDRESS	INSTRUCTION	OPERAND	COMMENT
200	Load I/O	5	Initiate keyboard read
201	Test I/O	5	Check for completion
	Branch Not Ready	201	Loop until complete
	In	5	Load data byte

(b) Isolated I/O

Programmed I/O Summary

- Main problems
 - Processor waits a long time for I/O module to be ready for receive/transmit data
 - Processor waits and repeatedly checks status of I/O module
 - Performance is severely degraded

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Interrupt-Driven I/O

- Overcomes CPU waiting
- No repeated CPU checking of device
- CPU issues I/O command to a module and goes on to do other useful processing
- I/O module interrupts when ready

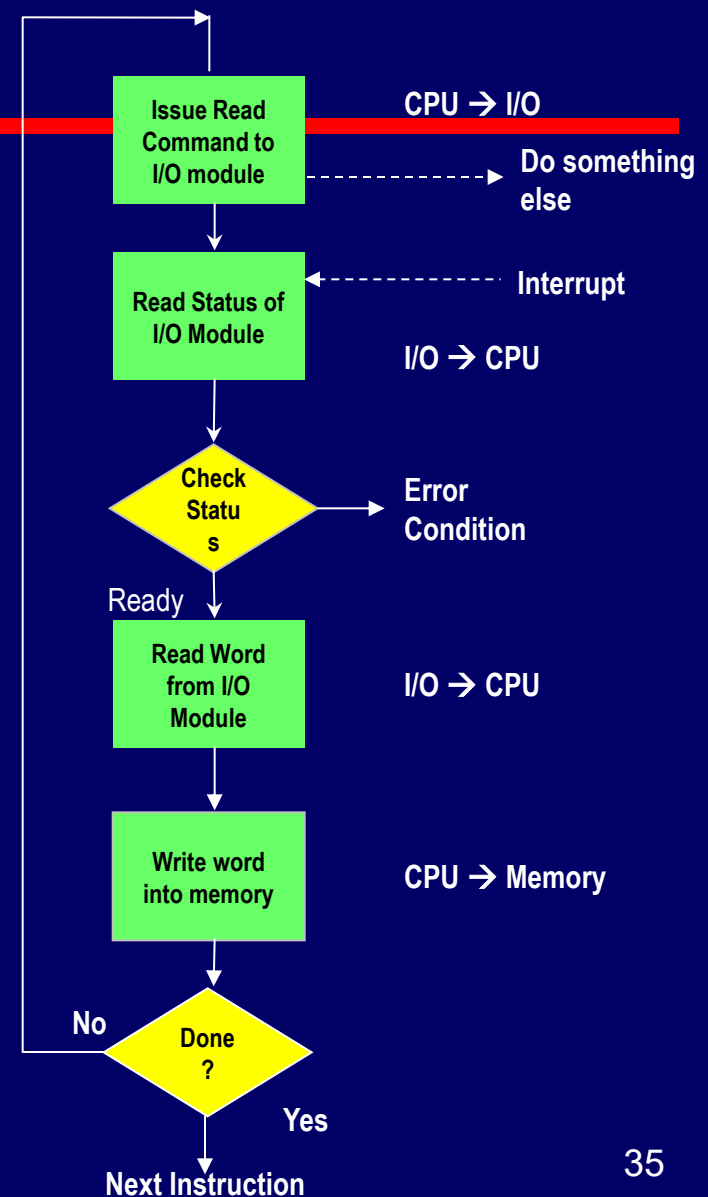
I/O Module Viewpoint – for Input

- CPU issues read command
- I/O module gets data from peripheral whilst CPU does other work
- When data is in I/O module's data register, I/O module interrupts CPU
 - Via control line
- I/O module waits until data is requested by CPU
- CPU requests data
- I/O module transfers data
 - Via data line
- I/O module ready for another I/O operation

CPU Viewpoint – for Input

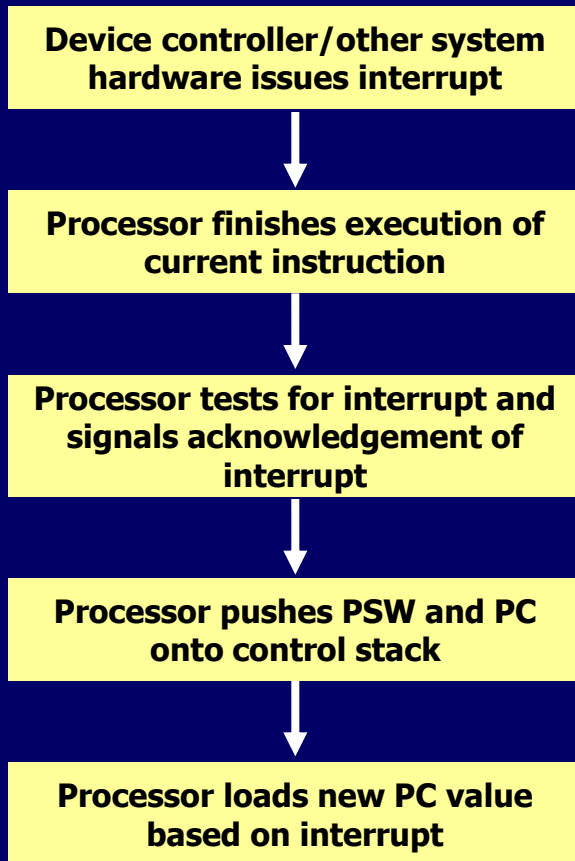
- Issue read command
- Do other work
- Check for interrupt at end of each instruction cycle
- If interrupted:
 - Save context (registers, e.g. PC, etc.)
 - Process interrupt
 - Fetch data (from I/O module) & store in memory
- Restores context of program and resumes execution

Interrupt-Driven I/O - for Input

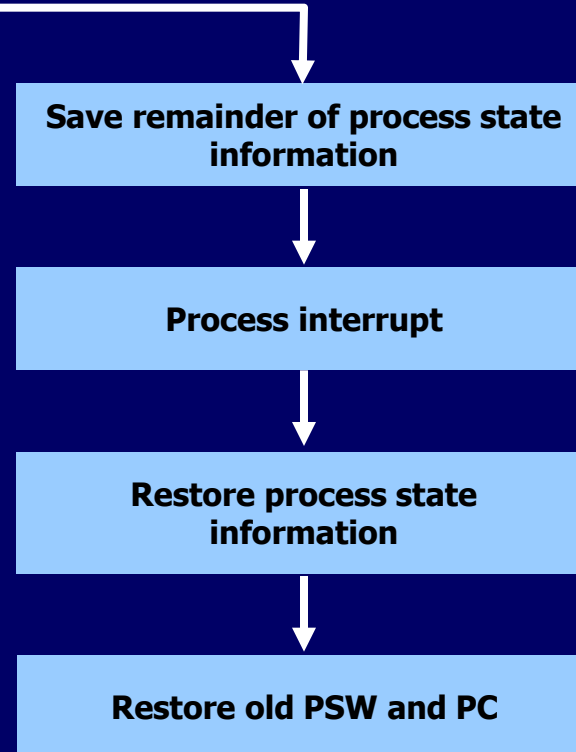


Interrupt Processing

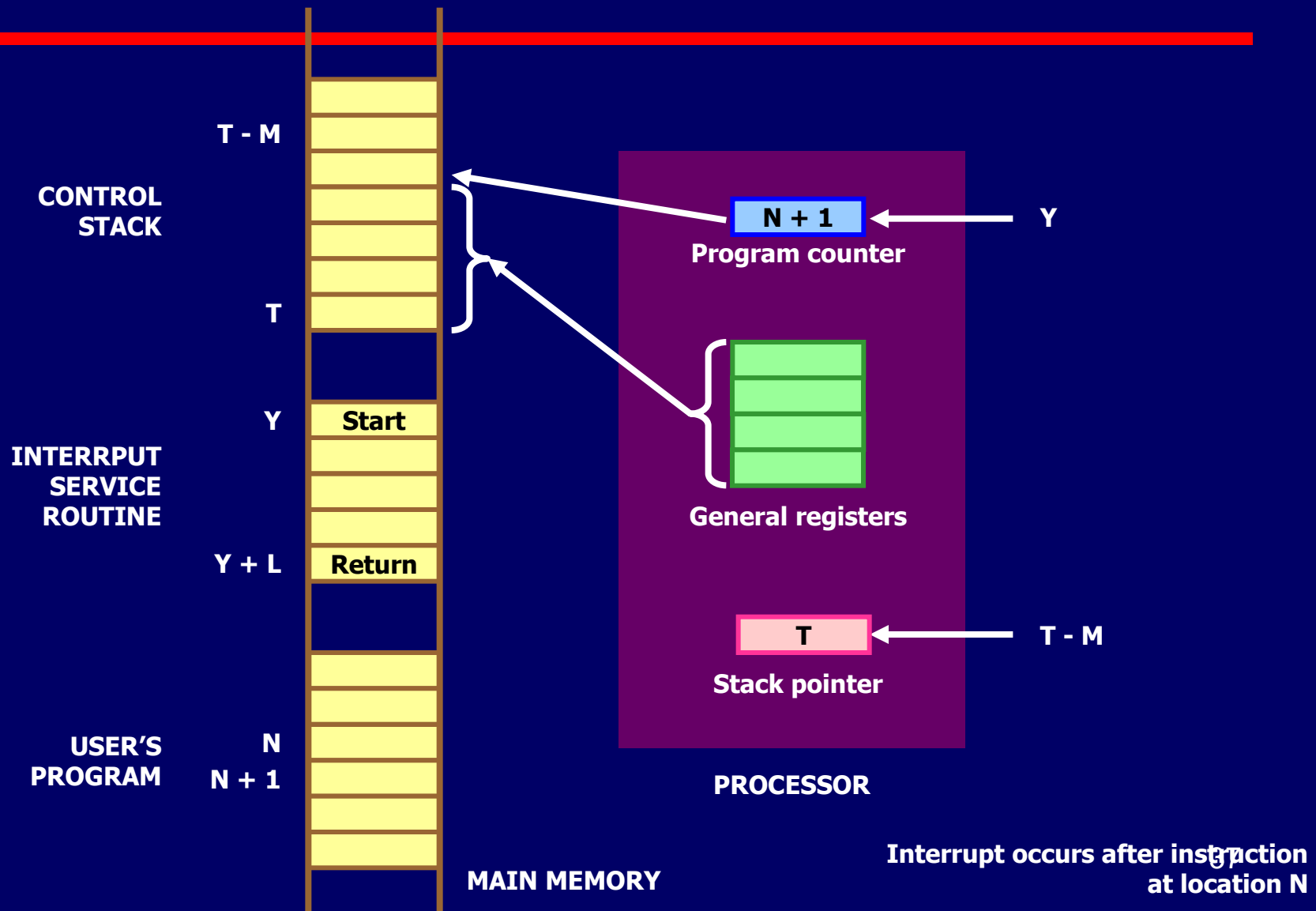
HARDWARE



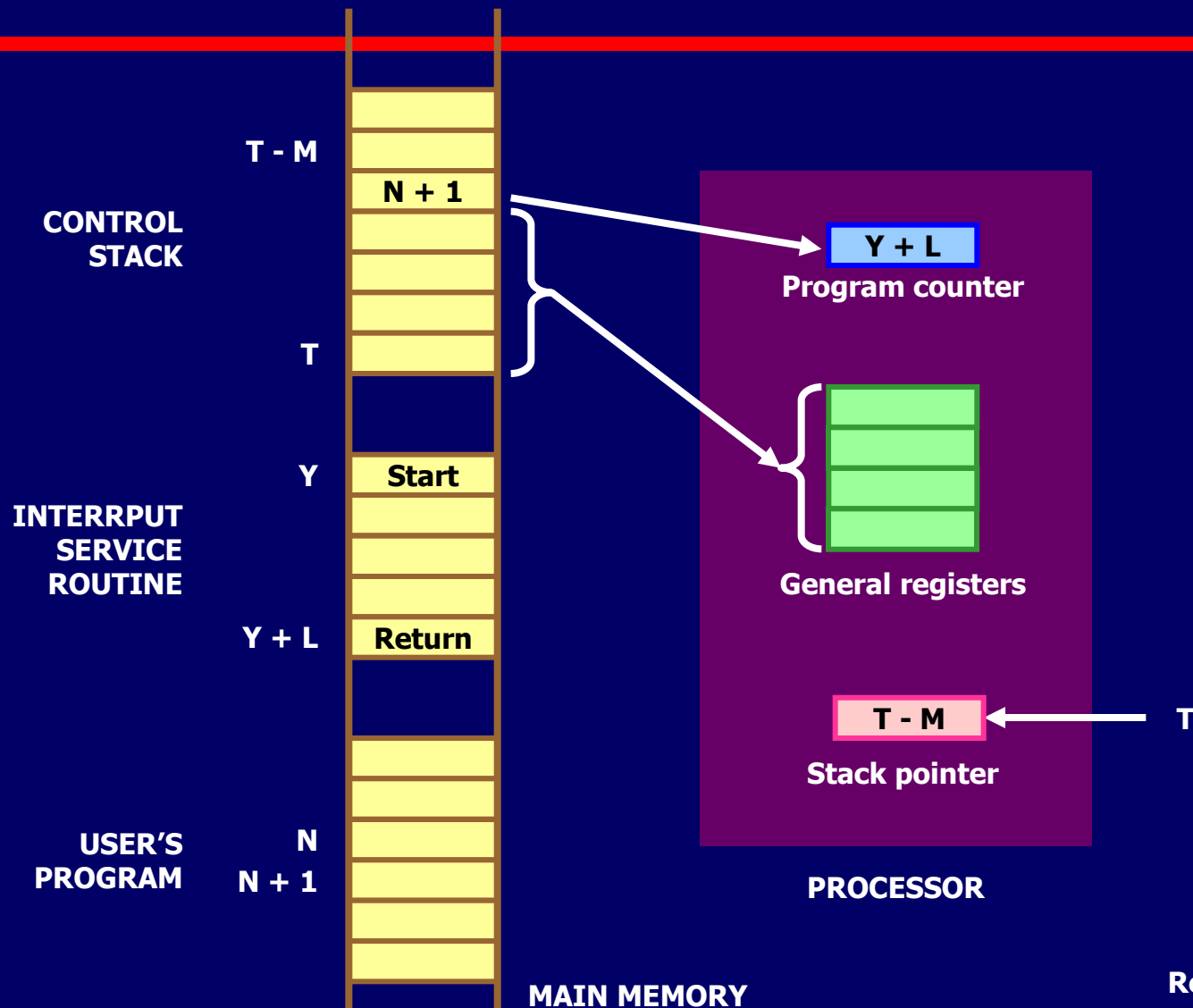
SOFTWARE



Interrupt Processing



Interrupt Processing



Design Issues

- How do you identify the module issuing the interrupt?
- How do you deal with multiple interrupts? Which interrupt to process first?
 - i.e. an interrupt handler being interrupted

Identifying Interrupting Module

- Different line for each module
 - i.e. different processor pins for different devices (too many is impractical)
 - Limits number of devices
- Software poll
 - Branches to interrupt service routine and asks each module in turn
 - E.g. TESTI/O (with I/O module address) – I/O module responds positively if it set the interrupt
 - Or each I/O module has addressable status register – processor read status register to identify interrupting module
 - Slow

Identifying Interrupting Module

- Daisy Chain or Hardware poll
 - Interrupt Acknowledge sent down a chain
 - Module responsible places vector (address/identifier) on data bus
 - CPU uses vector to identify handler routine
 - Avoids need to execute a general interrupt service routine first
 - Known as vectored interrupt
- Bus Master / Bus Arbitration
 - Module must claim control of the bus before it can raise interrupt
 - Only one module can raise interrupt line at a time
 - Processor acknowledges interrupt on acknowledge line and module places vector on data lines
 - e.g. PCI & SCSI

Multiple Interrupts

- Each interrupt line has a priority
- In software polling / daisy chain
 - Order of polling determines priority
- Higher priority lines can interrupt lower priority lines
- If bus mastering only current master can interrupt

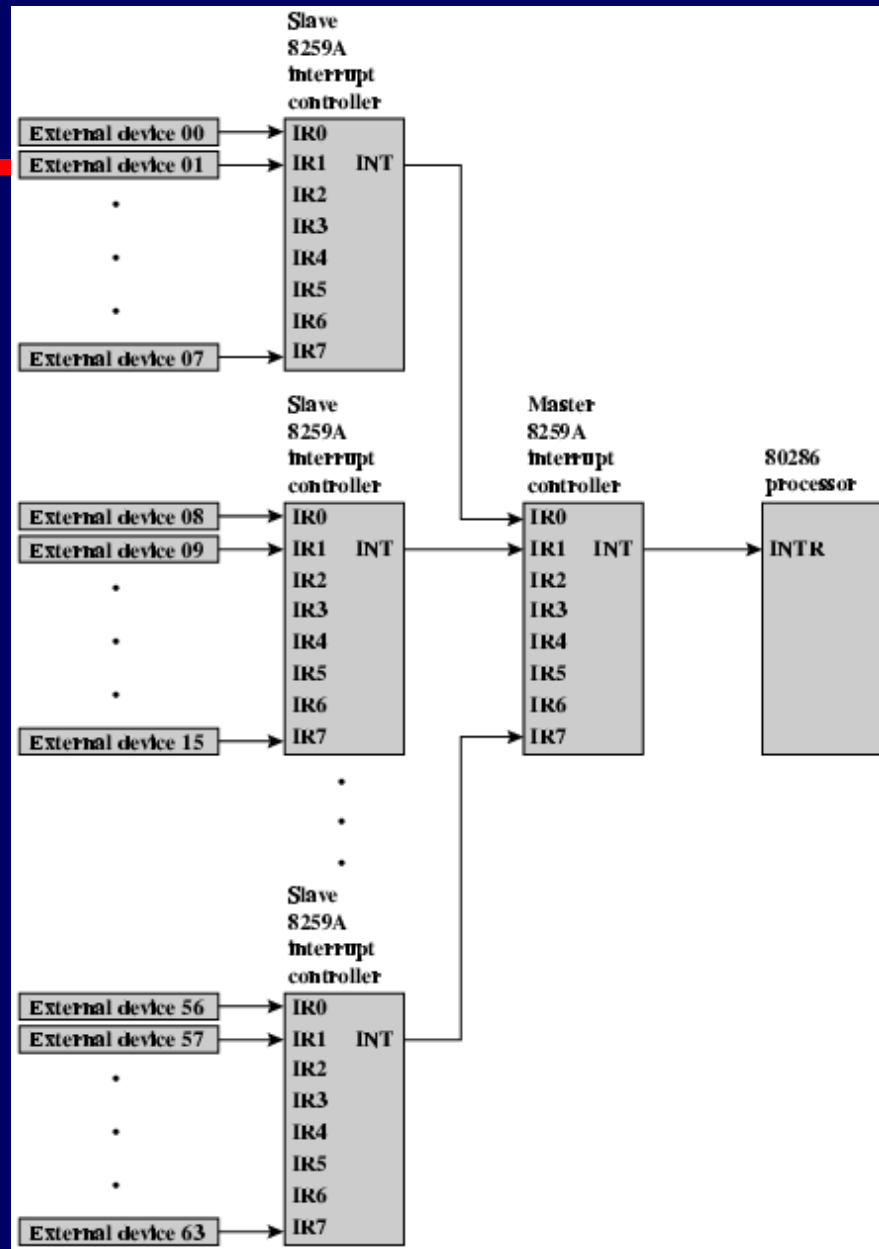
Example - PC Bus

- 80x86 has one interrupt line
- 8086 based systems use one 82C59A interrupt controller
- 82C59A has 8 interrupt lines

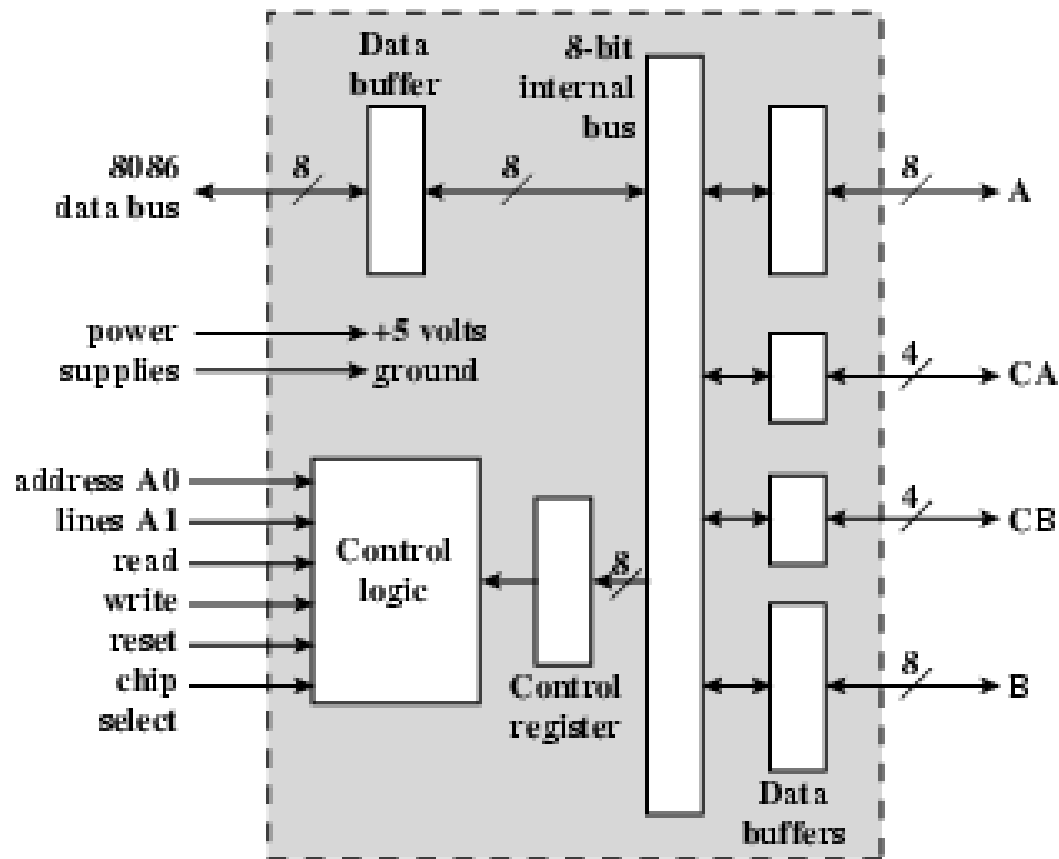
Sequence of Events

- 82C59A accepts interrupts
- 82C59A determines priority
- 82C59A signals 8086 (raises INTR line)
- CPU Acknowledges
- 82C59A puts correct vector on data bus
- CPU processes interrupt

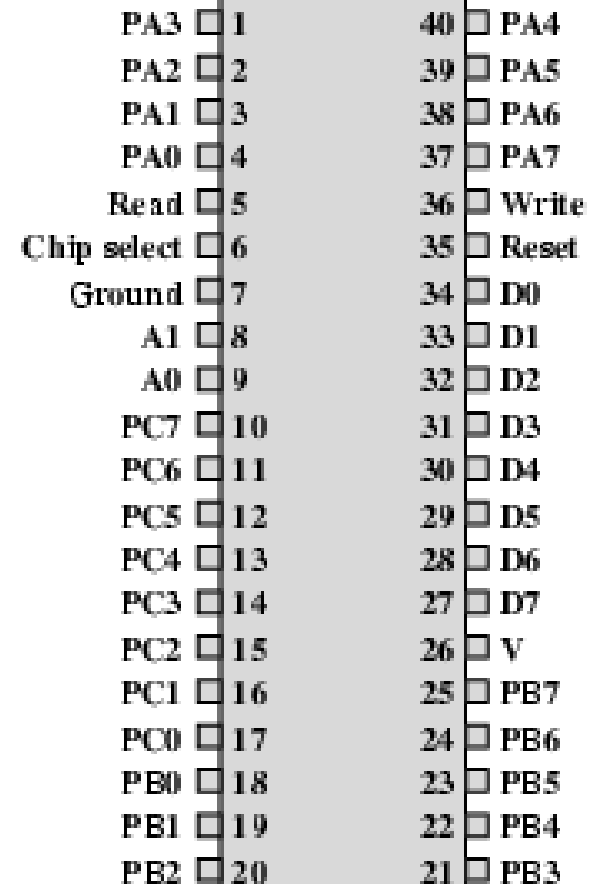
82C59A Interrupt Controller



Intel 82C55A Programmable Peripheral Interface

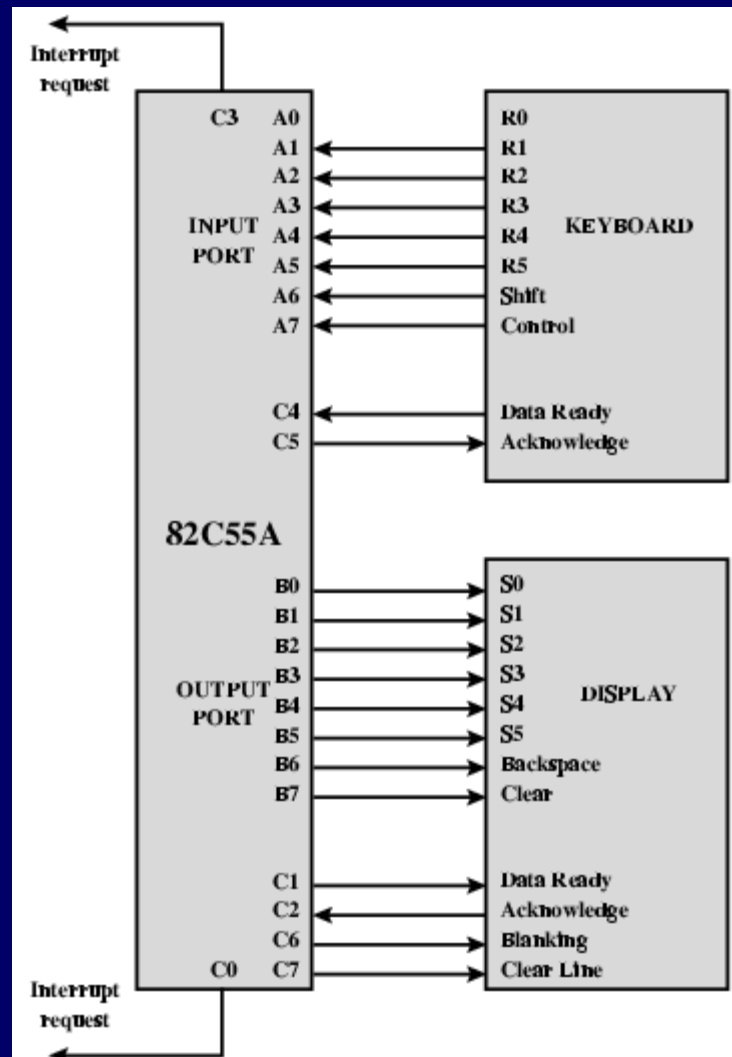


(a) Block diagram



(b) Pin layout

Using 82C55A To Control Keyboard/Display



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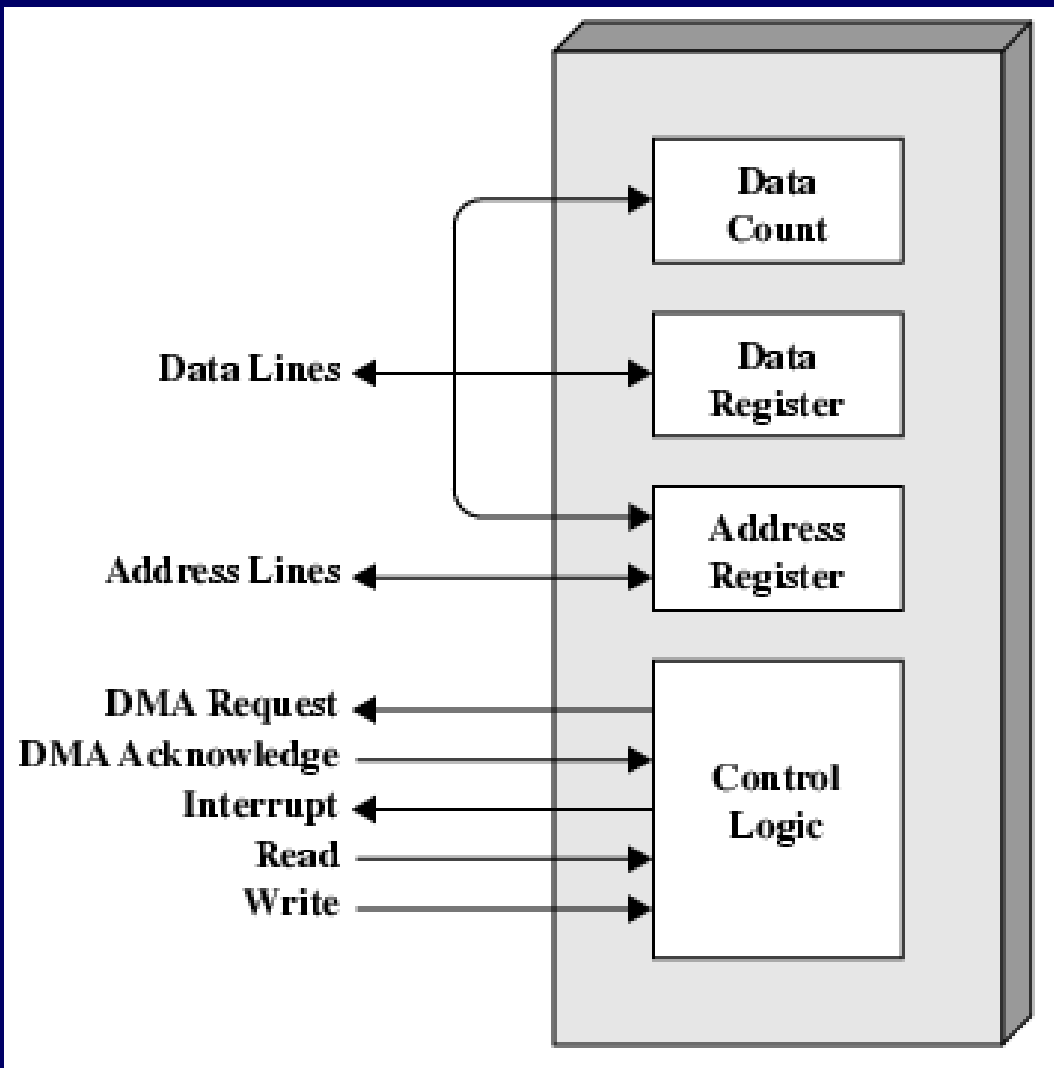
Direct Memory Access

- Drawbacks of Interrupt driven and programmed I/O
 - Interrupt driven and programmed I/O require active CPU intervention
 - Transfer rate is limited by the speed the processor can test and service a device
 - CPU is tied up managing the I/O transfer
- DMA is the answer

DMA Function

- Additional Module (hardware) on bus
- DMA controller takes over from CPU for I/O
 - When CPU does not need the bus
 - Or forces CPU to suspend operation temporarily
 - More common – cycle stealing

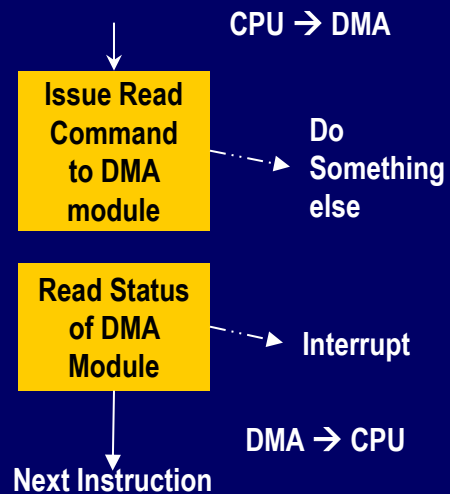
DMA Module Diagram



DMA Operation

- CPU tells DMA controller:
 - Read/Write
 - Device address
 - Starting address of memory block for data
 - Amount of data to be transferred
- CPU carries on with other work
- DMA controller deals with transfer
 - CPU not involved
- DMA controller sends interrupt when finished
 - CPU involved at the beginning and at the end

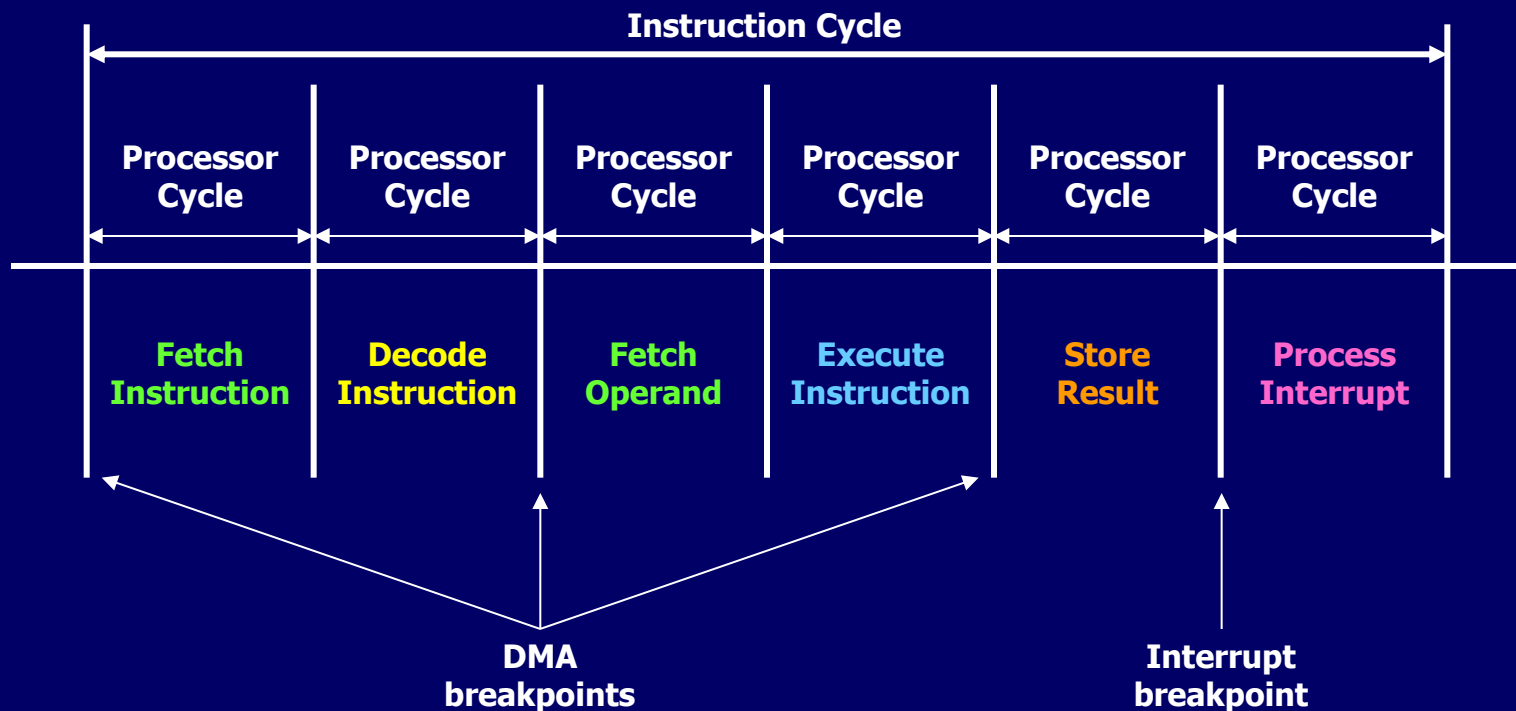
DMA Operation



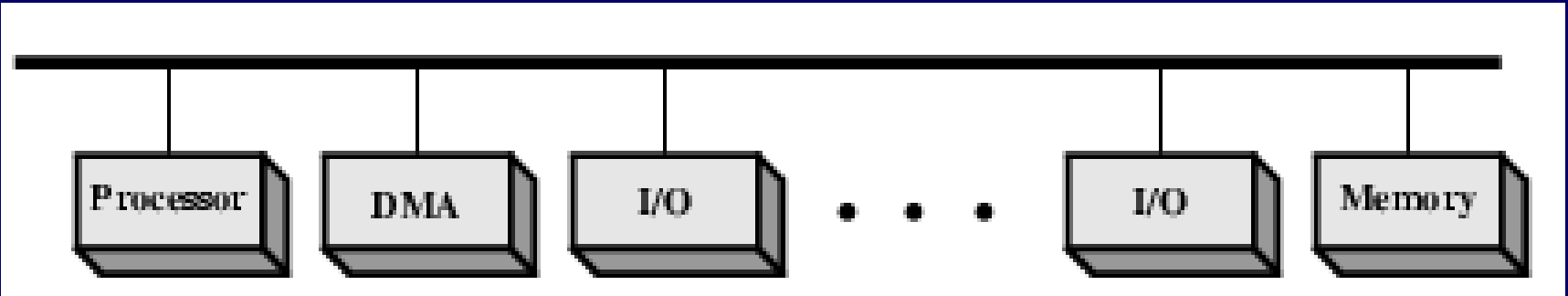
DMA Transfer - Cycle Stealing

- DMA controller takes over bus for a cycle
- Transfer of one word of data
- Not an interrupt
 - CPU does not switch/save context
- CPU suspended just before it accesses bus
 - i.e. before an operand or data fetch or a data write
- Slows down CPU but not as much as CPU doing transfer

DMA Transfer - Cycle Stealing

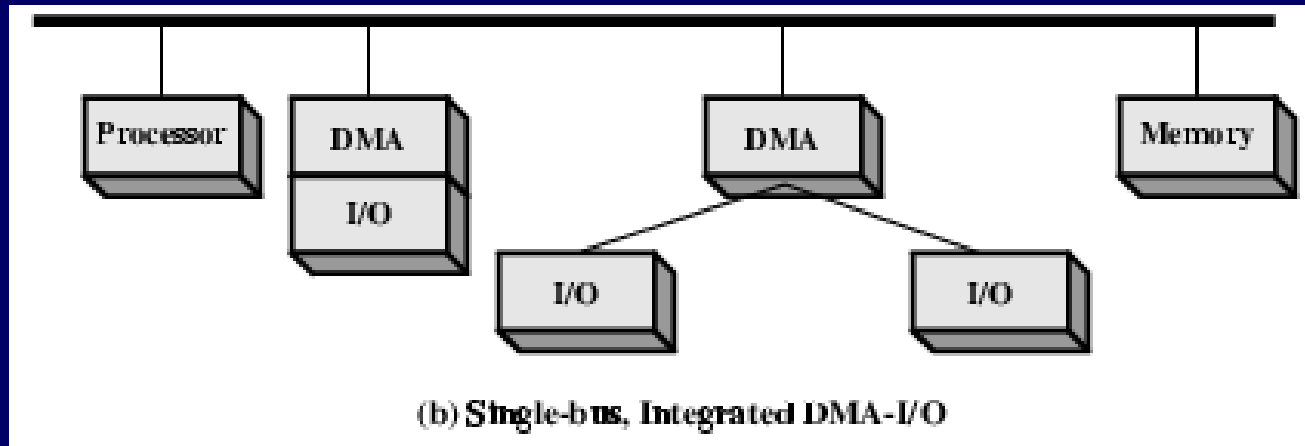


DMA Configurations



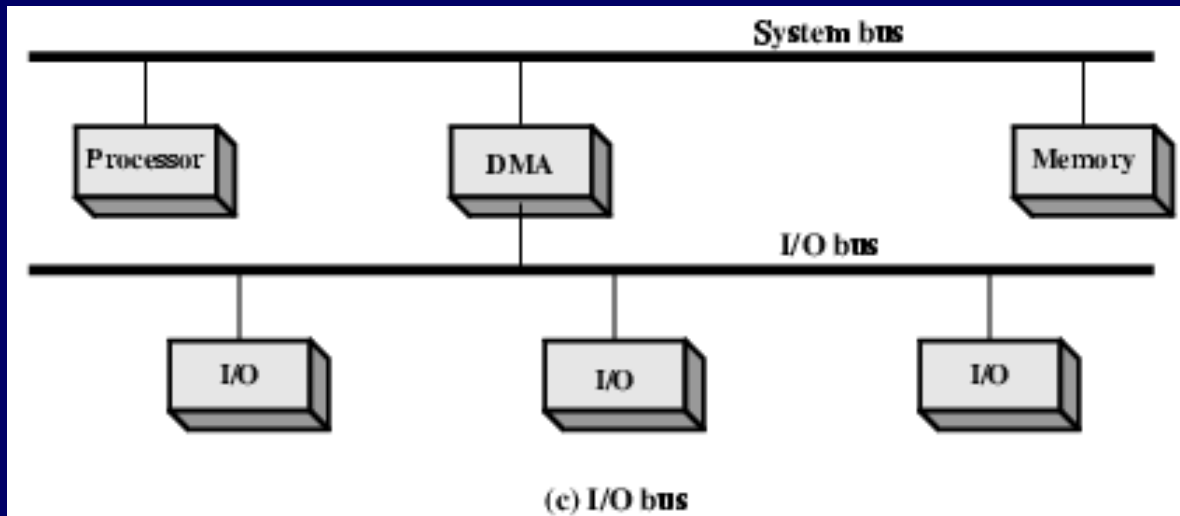
- Single Bus, detached DMA controller
- DMA controller uses programmed I/O to exchange data
- Each transfer uses bus twice
 - I/O to DMA
 - Then DMA to memory
- Inexpensive but inefficient
 - CPU is suspended twice

DMA Configurations



- Single Bus, integrated DMA controller
- Controller may support >1 device
- Each transfer uses bus once
 - DMA to memory
- CPU is suspended once

DMA Configurations



- Separate I/O Bus
- Bus supports all DMA enabled devices
- Each transfer uses bus once
 - DMA to memory
- CPU is suspended once

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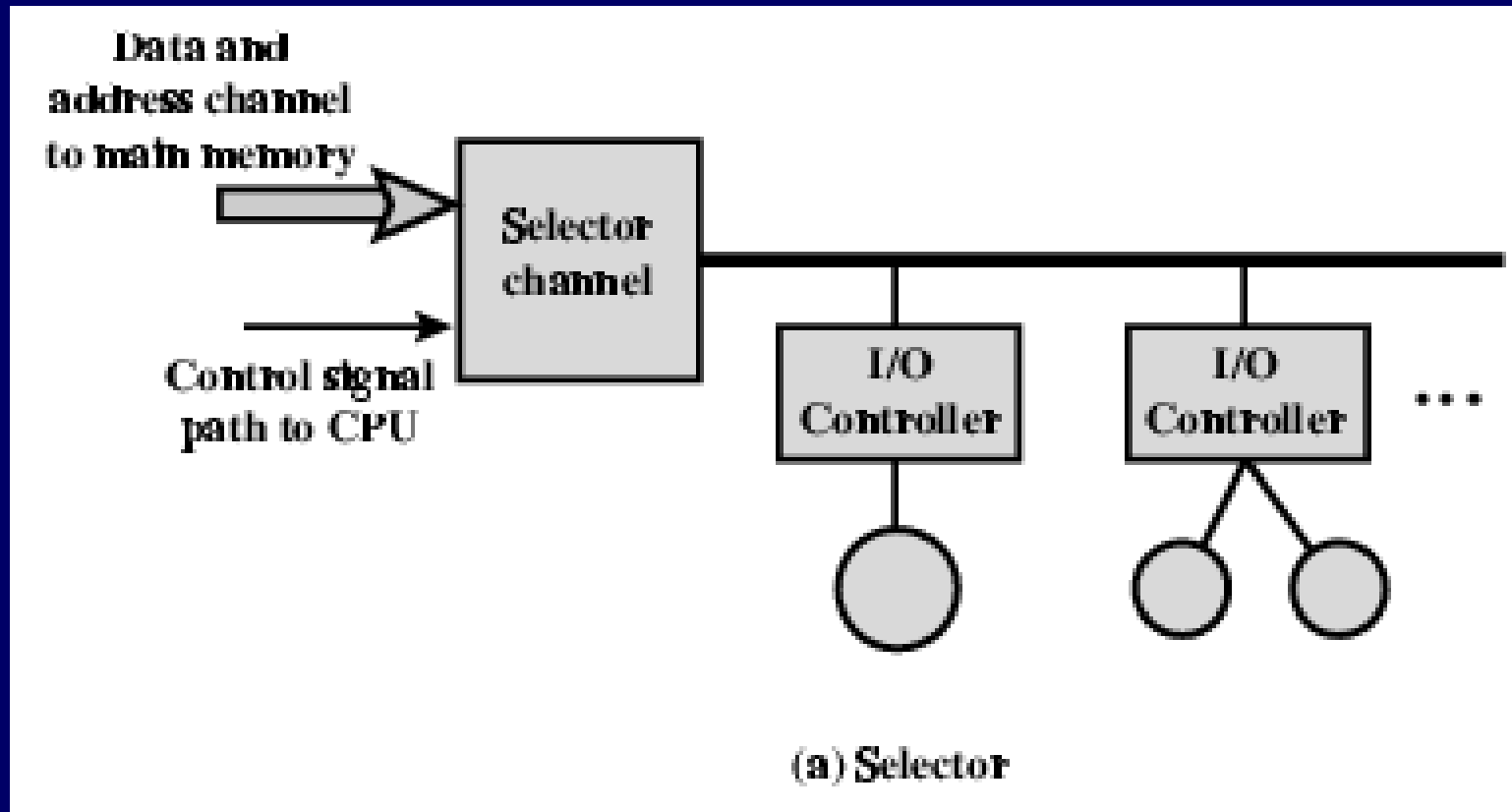
I/O Channels

- I/O devices getting more sophisticated
- e.g. 3D graphics cards
- CPU instructs I/O controller to do transfer
- I/O controller does entire transfer
- Improves speed
 - Takes load off CPU
 - Dedicated processor is faster

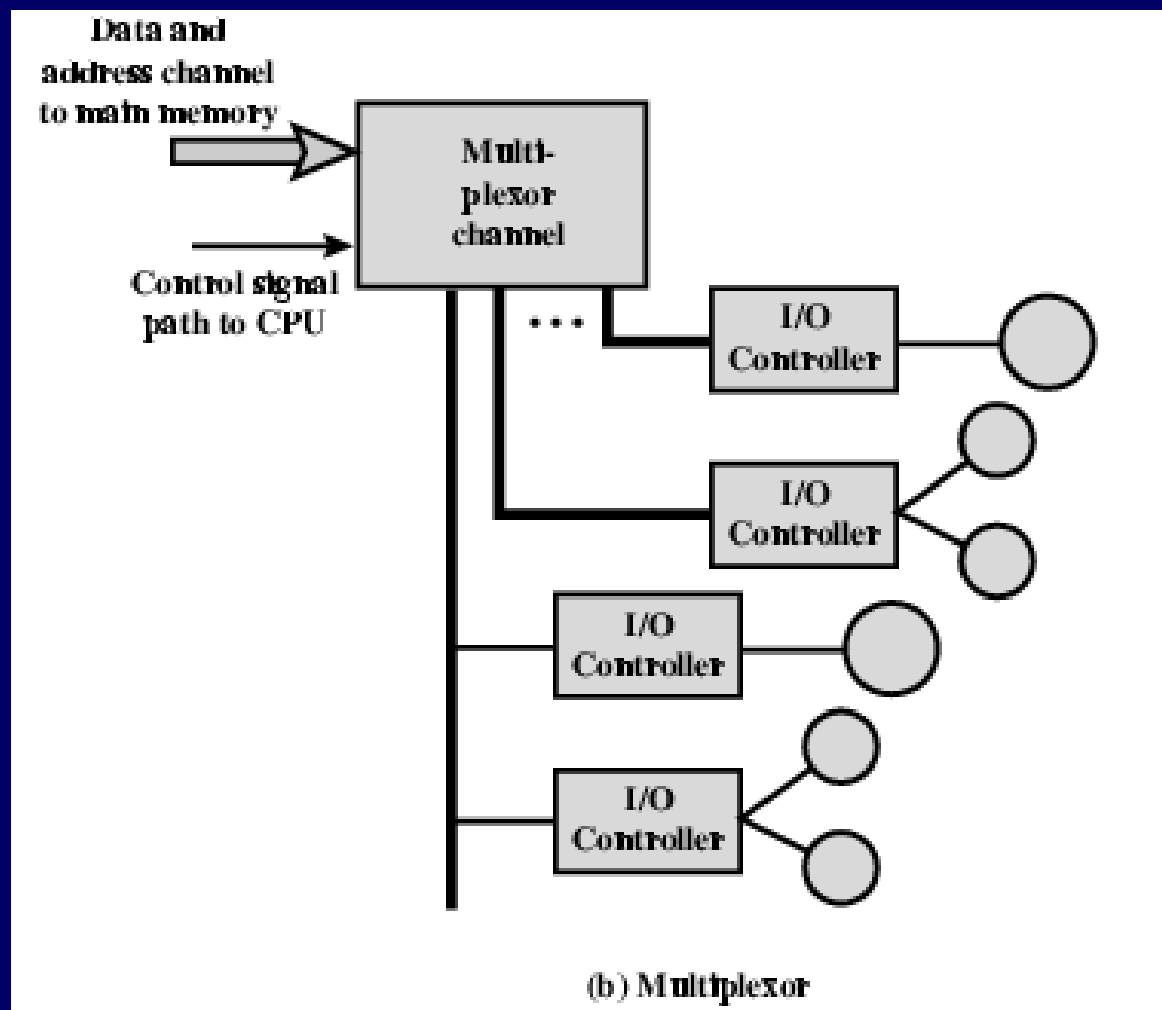
I/O Channels

- Extension of DMA concept
- CPU does not execute I/O instructions
 - CPU instructs I/O channel to execute I/O instructions from memory (I/O program)
 - I/O program specifies devices, storage areas, priority, and actions to be taken for error conditions
- 2 types
 - Selector channel
 - Controls multiple devices but selects only one for data transfer at any one time
 - Multiplexor channel
 - Handles multiple devices at the same time

I/O Channel Architecture



I/O Channel Architecture

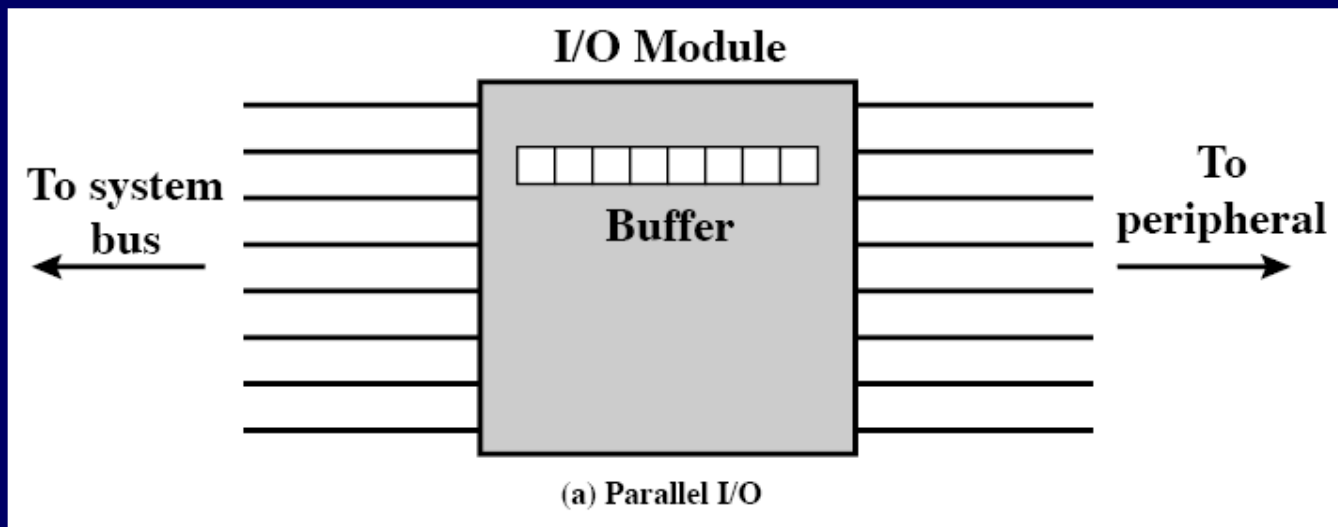


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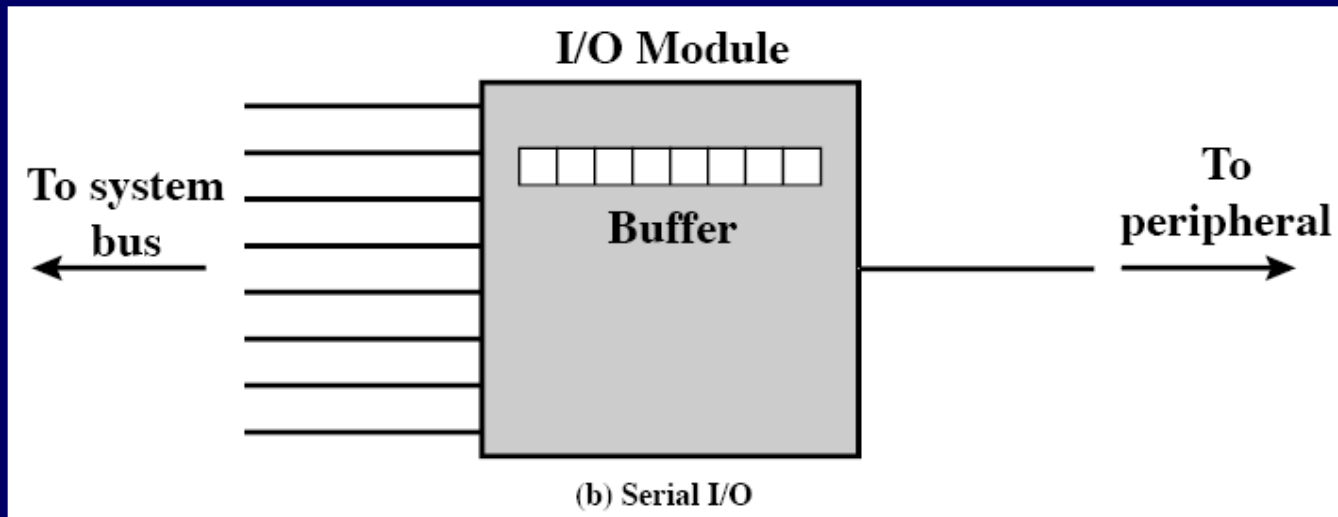
Interfacing

- Connecting devices together (from I/O module)
- Parallel interface
 - Multiple lines connecting I/O module and peripheral
 - Multiple bits transferred simultaneously
 - Higher speed peripherals, e.g. disks, printer



Interfacing

- Serial interface
 - Only one line used to transmit data
 - Bits transmitted one at a time
 - Slower peripherals, e.g. terminals, mice, joystick



- E.g. FireWire, USB

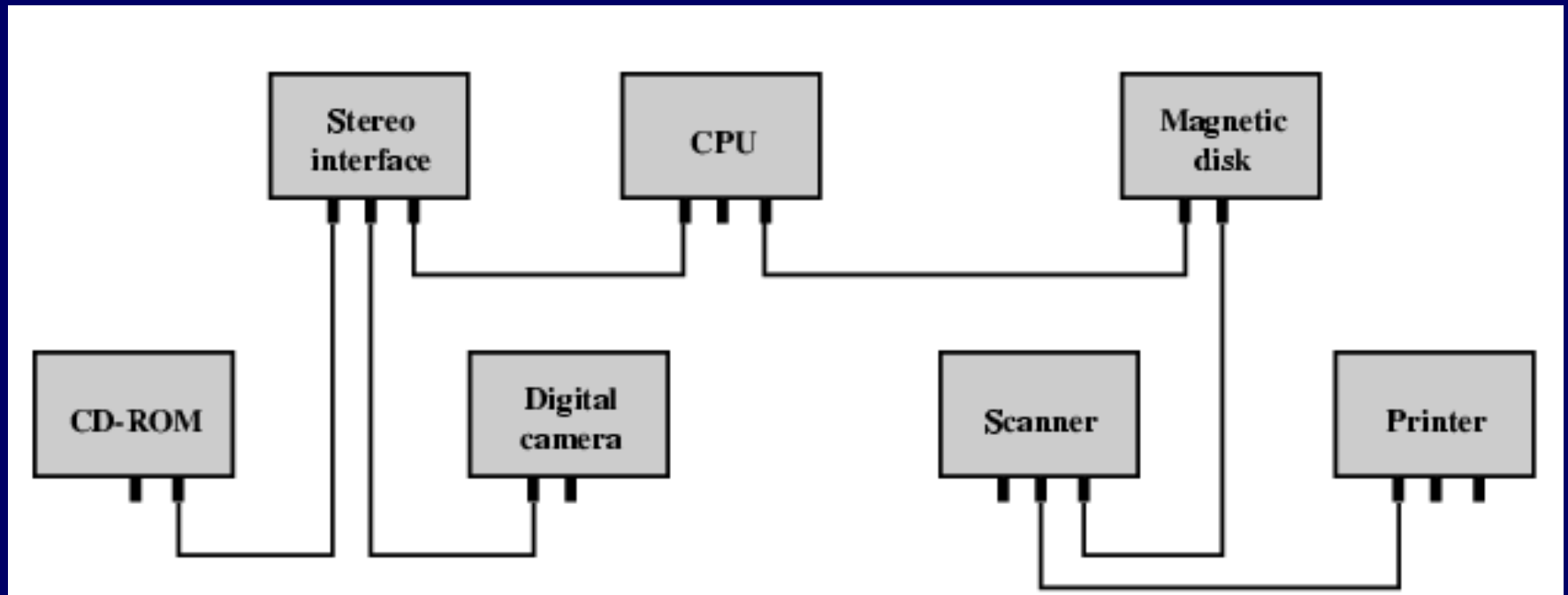
IEEE 1394 FireWire

- High performance serial bus
- Fast
- Low cost
- Easy to implement
- Also being used in digital cameras, VCRs and TV

FireWire Configuration

- Daisy chain
- Up to 63 devices on single port
 - Really 64 of which one is the interface itself
- Up to 1022 buses can be connected with bridges
- Automatic configuration
- No bus terminators
- May be tree structure

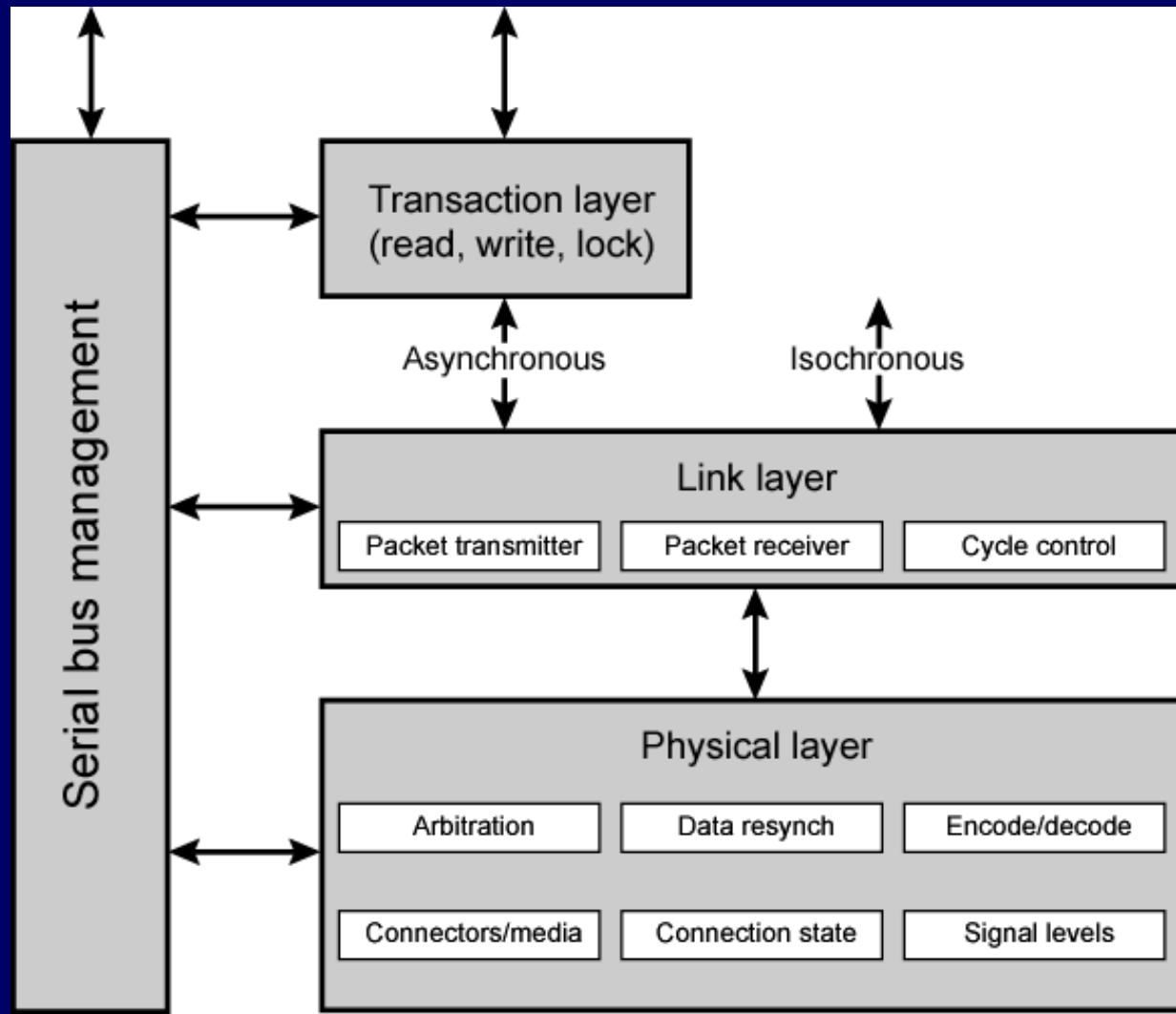
Simple FireWire Configuration



FireWire 3 Layer Stack

- Physical
 - Transmission medium, electrical and signaling characteristics
- Link
 - Transmission of data in packets
- Transaction
 - Request-response protocol

FireWire Protocol Stack



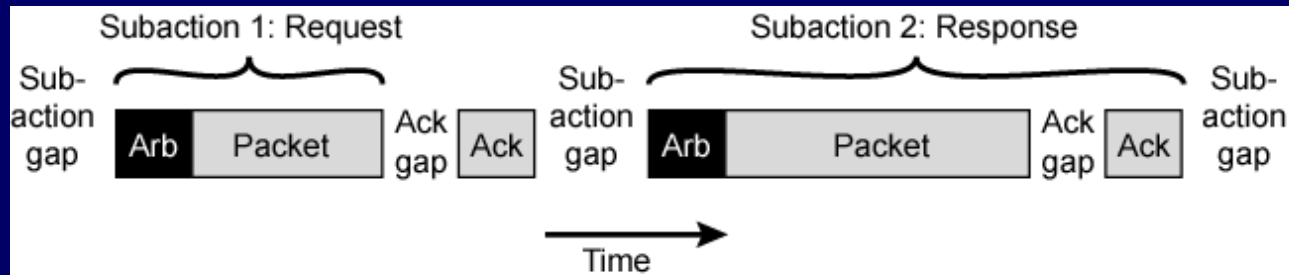
FireWire - Physical Layer

- Data rates from 25 to 400Mbps
- Two forms of arbitration
 - Based on tree structure
 - Root acts as arbiter
 - First come first served
 - Natural priority controls simultaneous requests
 - i.e. who is nearest to root
 - Fair arbitration
 - Urgent arbitration

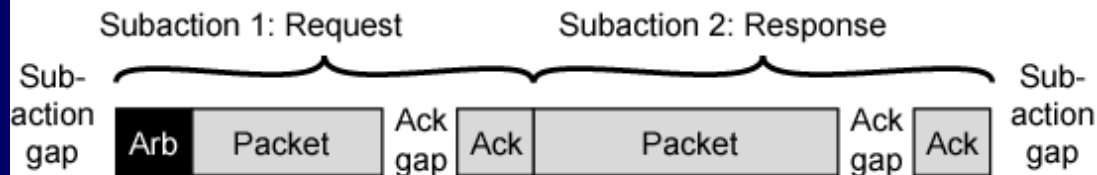
FireWire - Link Layer

- Two transmission types
 - Asynchronous
 - Variable amount of data and several bytes of transaction data transferred as a packet
 - To explicit address
 - Acknowledgement returned
 - Isochronous
 - Variable amount of data in sequence of fixed size packets at regular intervals
 - Simplified addressing
 - No acknowledgement

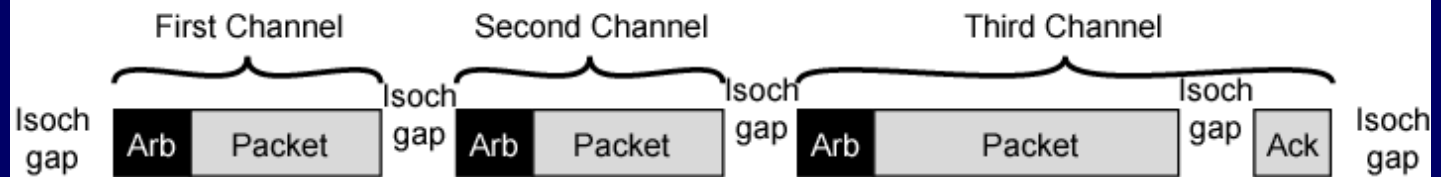
FireWire Subactions



(a) Example asynchronous subaction



(b) Concatenated asynchronous subactions



(c) Example isochronous subactions

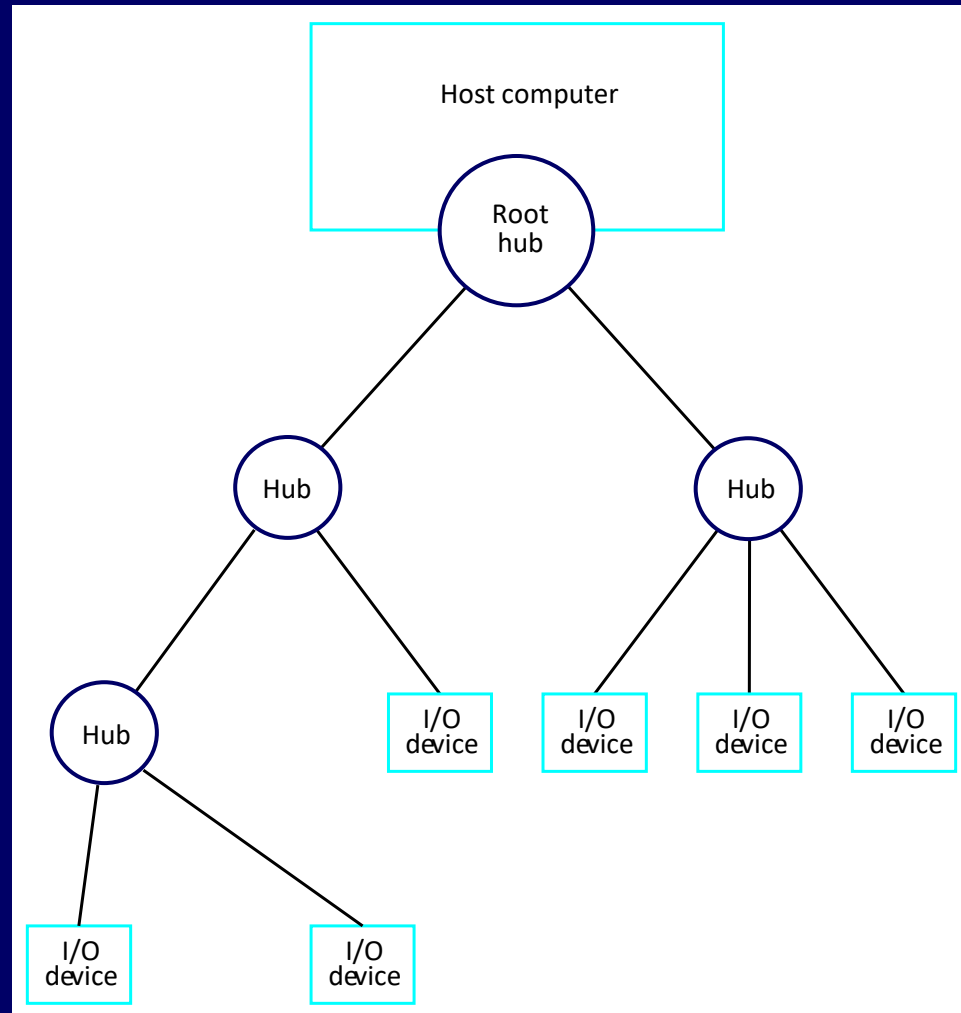
Universal Serial Bus (USB)

- Simple and low cost
- Overcomes difficulties in limited number of I/O ports
- Wide range of I/O devices
- Plug-and-play

Universal Serial Bus (USB) Configuration

- Tree structure
 - Accommodate large number of devices
 - Each node of tree has a hub device
 - Root hub connects tree to host computer
 - Leaves are I/O devices
 - Hub copies message that it receives from upstream connection to all downstream ports
 - Only addressed device will respond to message

Universal Serial Bus (USB) Configuration



Universal Serial Bus (USB)

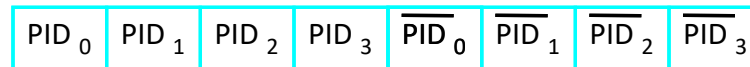
Addressing

- Each device on USB (hub or device) is assigned a 7-bit address
- When device is first connected to a hub, it has address 0
- Hub hardware is capable of detecting connected device
- Host polls hubs periodically to collect status information and learn about new devices or disconnected
- New devices assigned new unique USB address
- USB software must maintain complete picture of the bus topology and the connected devices at all times

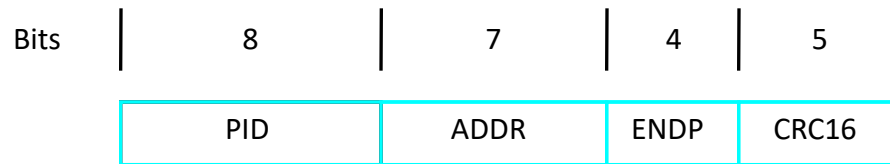
Universal Serial Bus (USB) Protocols

- Packet broad categories
 - Control: e.g. addressing a device to initiate data transfer, acknowledging data received, or indicating error
 - Data: information delivered to device
- Packet consists of one or more fields
 - First field is packet identifier (PID): 4 bits (16 packet types), transmitted twice – 2nd time complimented

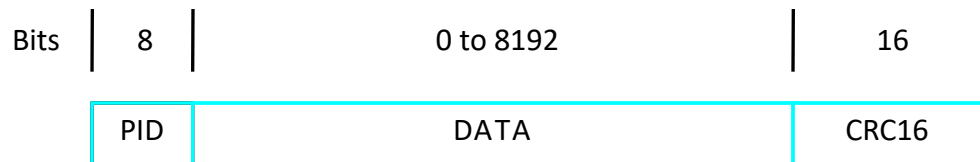
Universal Serial Bus (USB) Protocols



(a) Packet identifier field



(b) Token packet, IN or OUT (control packet)



(c) Data packet

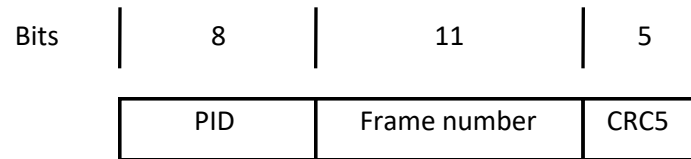
Universal Serial Bus (USB) Protocols



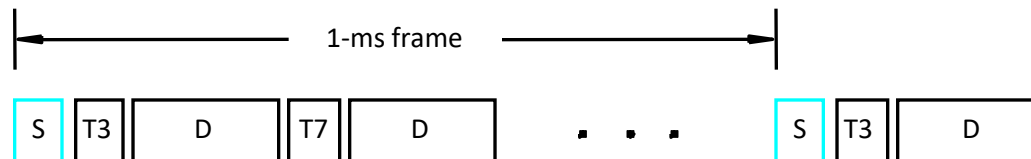
Universal Serial Bus (USB) Frames

- Supports isochronous data (data flows at set rate)
- 1 frame = 1 milliseconds long
 - Start of frame packet, SOF, (control packet) generated by root hub every 1 ms
 - SOF carries 11-bit frame number

Universal Serial Bus (USB) Frames



(a) SOF Packet



S - Start-of-frame packet

T n - Token packet, address = n

D - Data packet

A - ACK packet

(b) Frame example

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 - Programmed I/O, Interrupt-Driven I/O and Direct Memory Access (DMA)
- I/O Channels and Processors
- Interfaces
 - Parallel and serial interfaces
- Introduction to Networks
- RAID architectures

Introduction to Networks

- Network Interface Adaptor
 - Provides the link between a computer and the network
 - Requires a device driver to perform both data link and physical layer functions
 - Plugs into a bus slot or universal serial bus (USB) port on a computer
 - Also referred to as a network interface card (NIC)

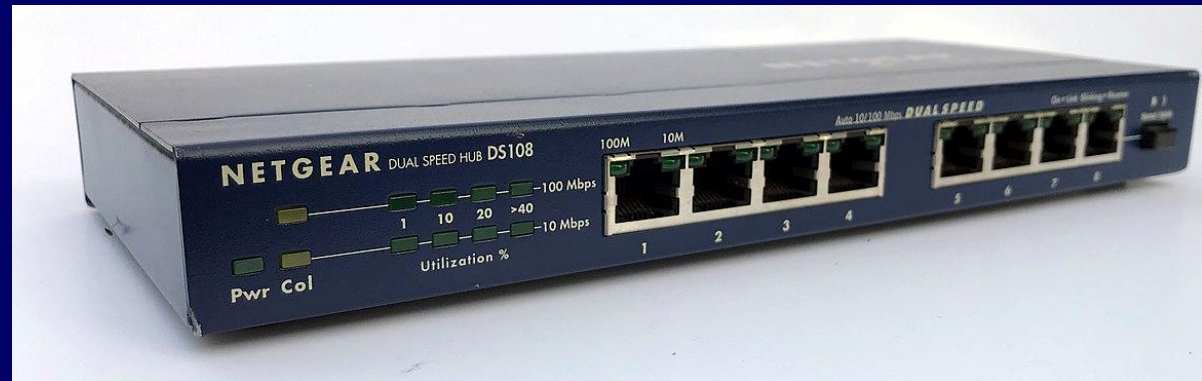


Introduction to Networks

- Network Interface Adaptor performs the following functions:
 - Data transfer, buffering, and encapsulation
 - Media Access Control (MAC)
 - Parallel/ serial conversion
 - Signal encoding and amplification

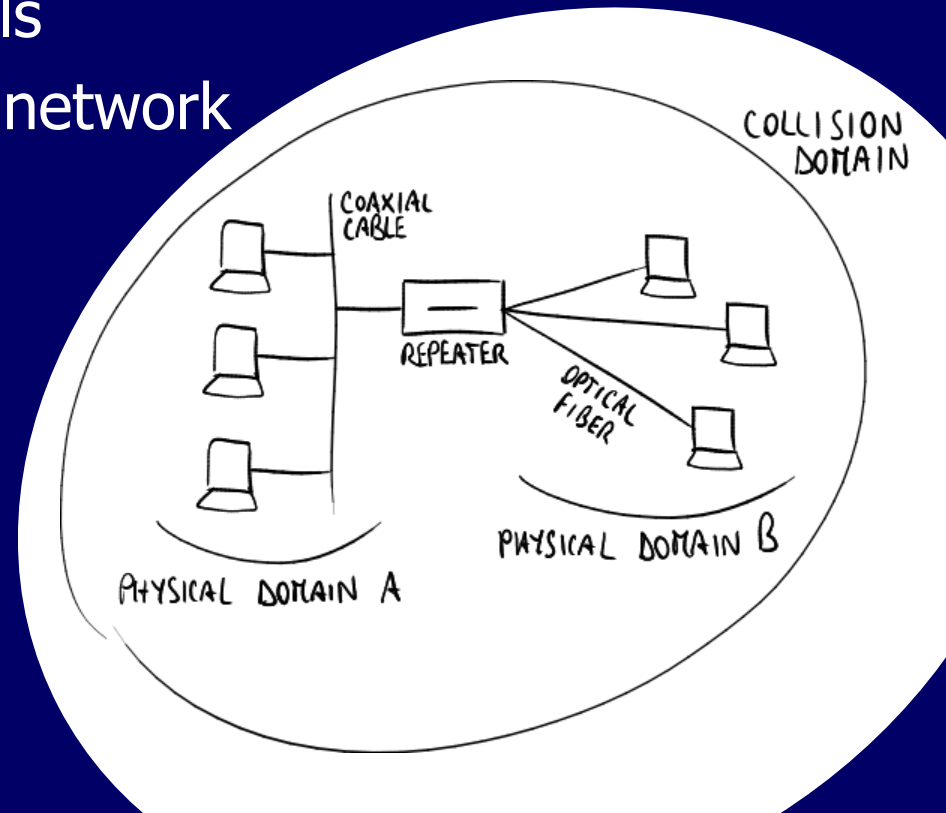
Introduction to Networks

- Hubs, repeaters, and concentrators are all physical layer devices that
 - Amplify and repeat signals
 - Extend the distance of a network



Introduction to Networks

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Introduction to Networks

- Hubs
 - 10Base-T and 100Base-TX/100Base-T4
 - Standards that define Ethernet networks that function at 10 Mbps and 100 Mbps
 - Uses baseband signaling over twisted pair wire

Introduction to Networks

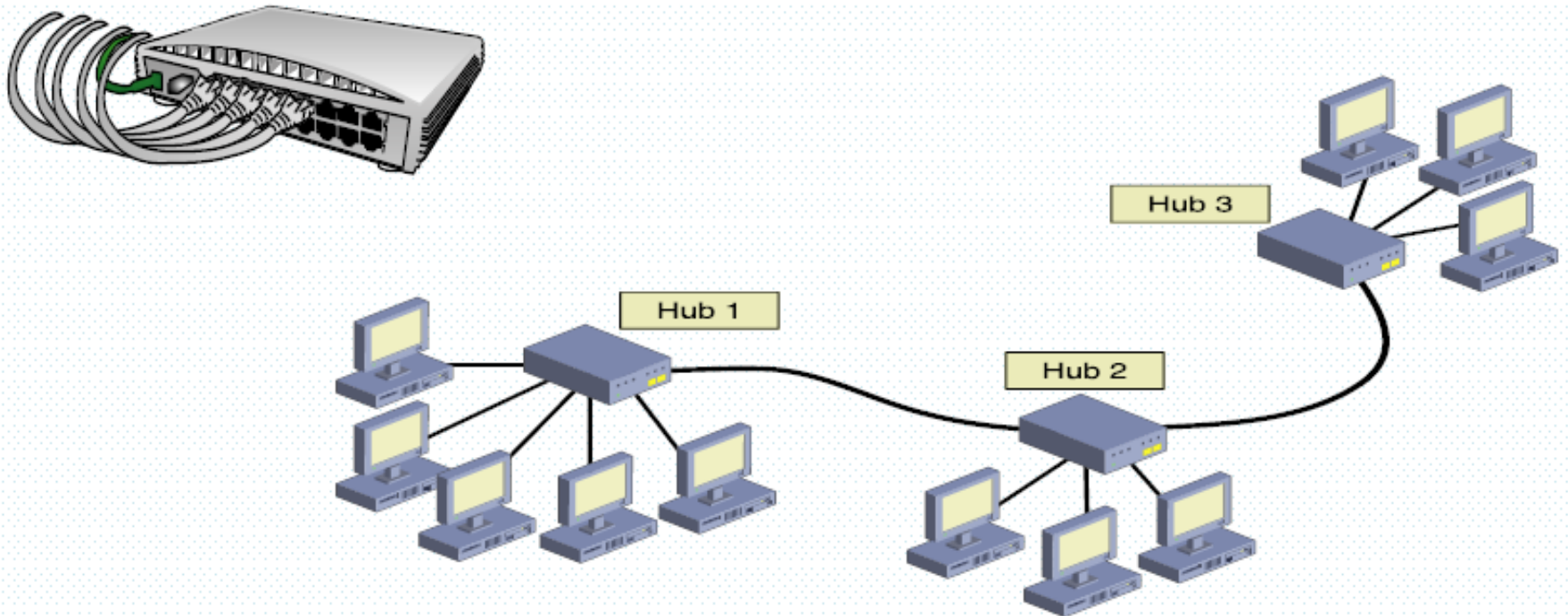
- 10Base-T
 - Maximum distance limitation for each connection: 100 meters, including workstation-to-hub and hub-to-hub connections
 - Can have up to four hubs connected to form a hierarchical star
 - Includes an internal crossover circuit
 - Uses an uplink port to form a hierarchical star

Introduction to Networks

- 100Base-TX and 100Base-T4
 - There are two types of hubs: Class I and Class II
 - The maximum distance for each node connection is 100 meters
 - Class II hub-to-hub connections can be no more than 5 meters long

Introduction to Networks

- 10Base-T hub

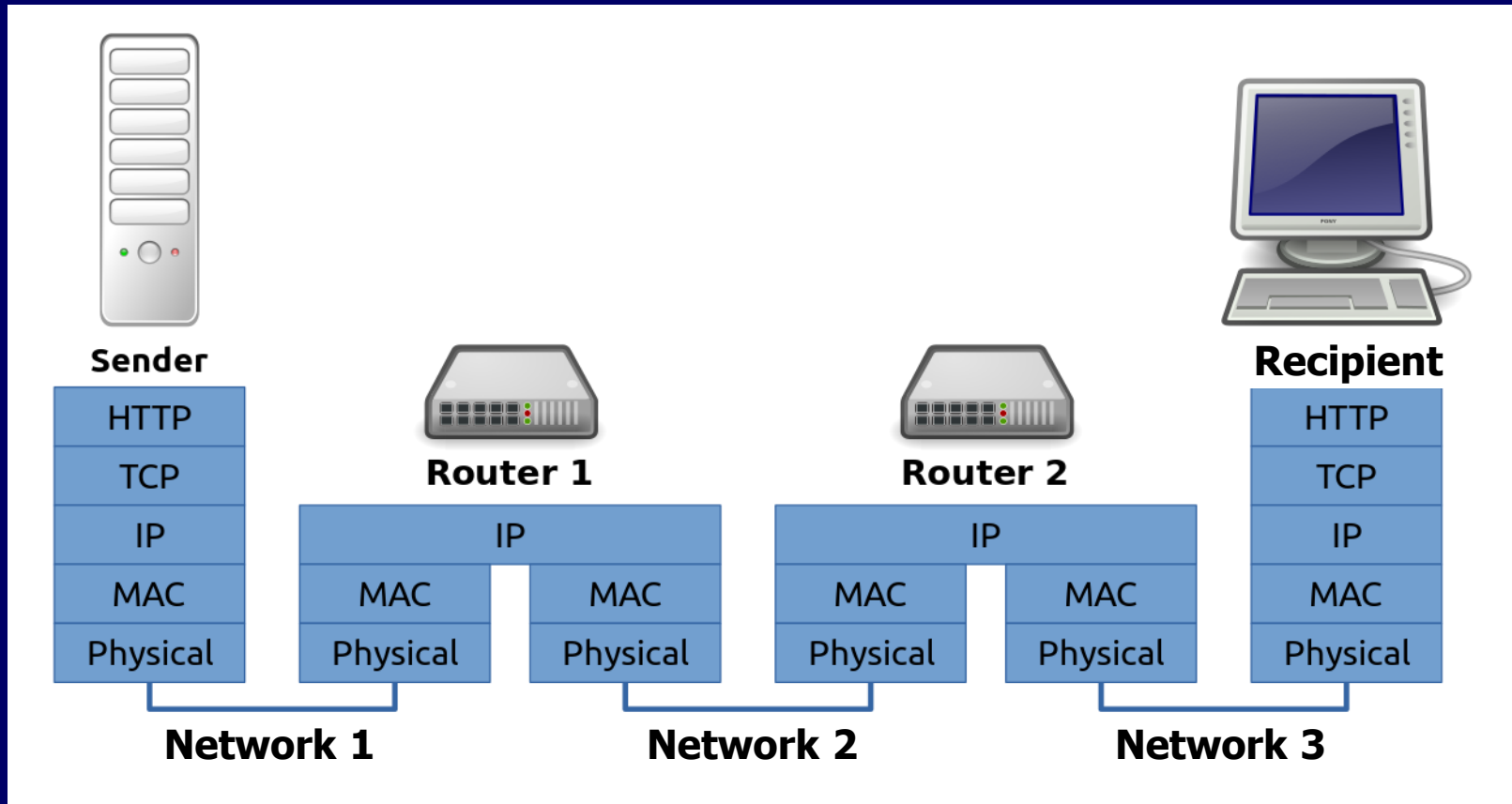


Introduction to Networks

- Router
 - Network layer devices that connects similar or different data link layer LANs
 - Must understand and support the network layer protocol and addressing
 - Perform fragmentation
 - Strip the data link header and footer off received frames
 - Add a new data link header and footer before transmitting frames
 - Use routing protocols to build routing tables and forward frames
 - Define separate broadcast domains

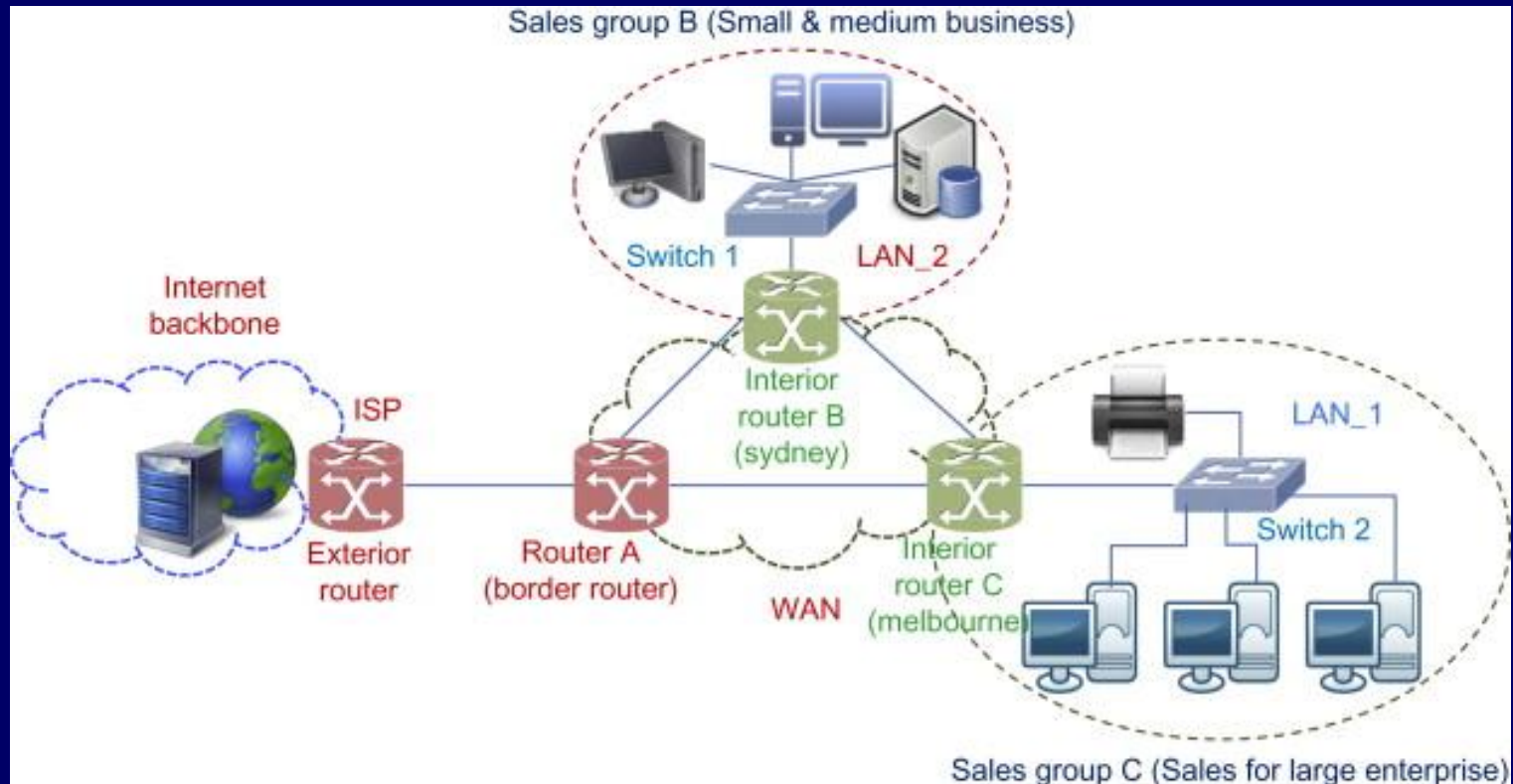
Introduction to Networks

- Router



Introduction to Networks

- Example of routed network



Introduction to Networks

- Gateway
 - Can include the functions of all seven layers of the OSI model
 - Connect dissimilar systems and protocols
 - Perform translation and conversion services

Outline

- External Devices
- I/O Modules
- I/O Operations
 - Programmed I/O, Interrupt-Driven I/O and Direct Memory Access (DMA)
- I/O Channels and Processors
- Interfaces
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RAID

Redundant Array of Independent Disks

- Consists of 7 levels
 - Levels do not imply a hierarchical relationship
- Different design architectures that share three common characteristics:
 1. Set of physical disk drives viewed by the operating system as a single logical drive
 2. Data are distributed across the physical drives of an array in a scheme known as striping
 3. Redundant disk capacity is used to store parity information, which guarantees data recoverability in case of a disk failure

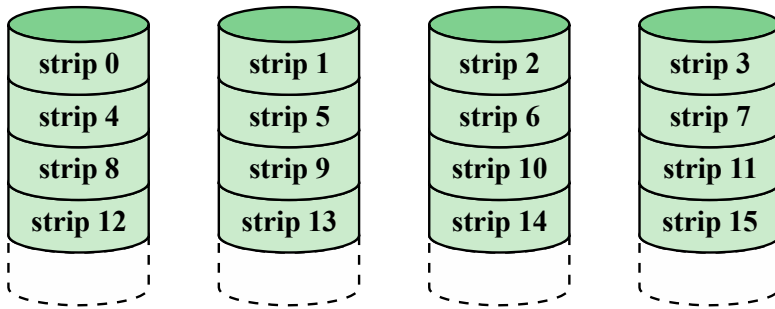
RAID Levels

Category	Level	Description	Disks Required	Data Availability	Large I/O Data Transfer Capacity	Small I/O Request Rate
Striping	0	Nonredundant	N	Lower than single disk	Very high	Very high for both read and write
Mirroring	1	Mirrored	$2N$	Higher than RAID 2, 3, 4, or 5; lower than RAID 6	Higher than single disk for read; similar to single disk for write	Up to twice that of a single disk for read; similar to single disk for write
Parallel access	2	Redundant via Hamming code	$N + m$	Much higher than single disk; comparable to RAID 3, 4, or 5	Highest of all listed alternatives	Approximately twice that of a single disk
	3	Bit-interleaved parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 4, or 5	Highest of all listed alternatives	Approximately twice that of a single disk
Independent access	4	Block-interleaved parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 3, or 5	Similar to RAID 0 for read; significantly lower than single disk for write	Similar to RAID 0 for read; significantly lower than single disk for write
	5	Block-interleaved distributed parity	$N + 1$	Much higher than single disk; comparable to RAID 2, 3, or 4	Similar to RAID 0 for read; lower than single disk for write	Similar to RAID 0 for read; generally lower than single disk for write
	6	Block-interleaved dual distributed parity	$N + 2$	Highest of all listed alternatives	Similar to RAID 0 for read; lower than RAID 5 for write	Similar to RAID 0 for read; significantly lower than RAID 5 for write

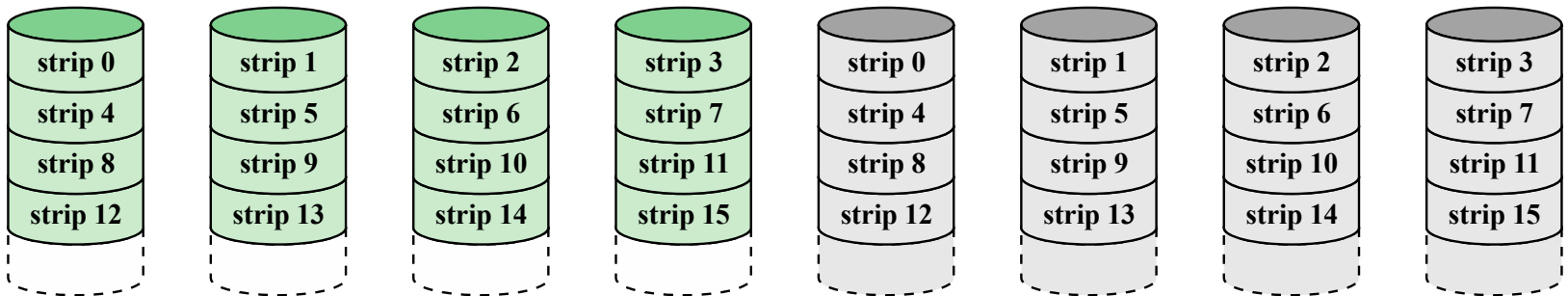
N = number of data disks; m proportional to $\log N$

RAID

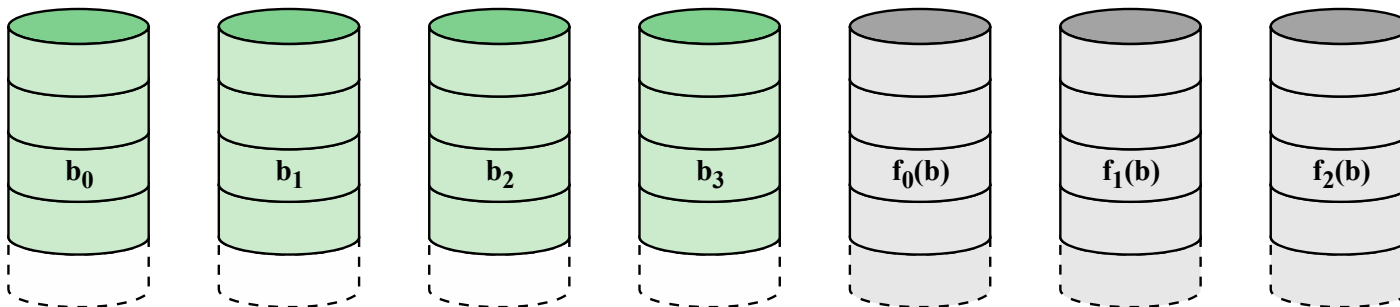
Level 0, 1, 2



(a) RAID 0 (non-redundant)

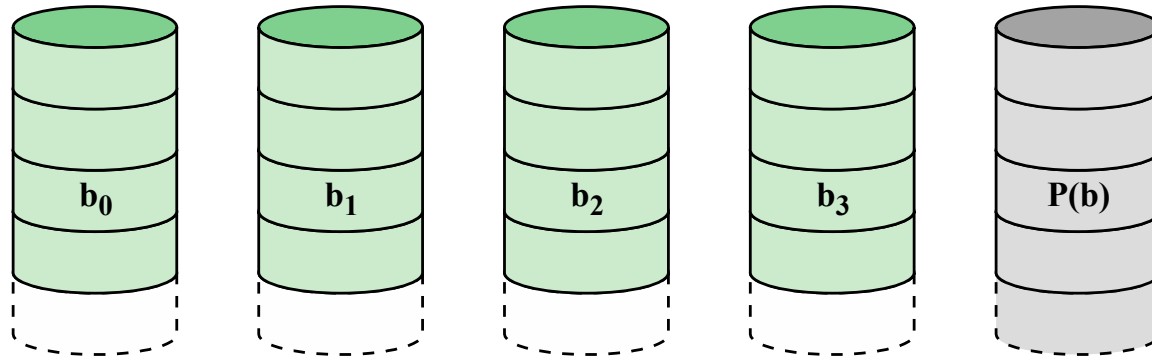


(b) RAID 1 (mirrored)

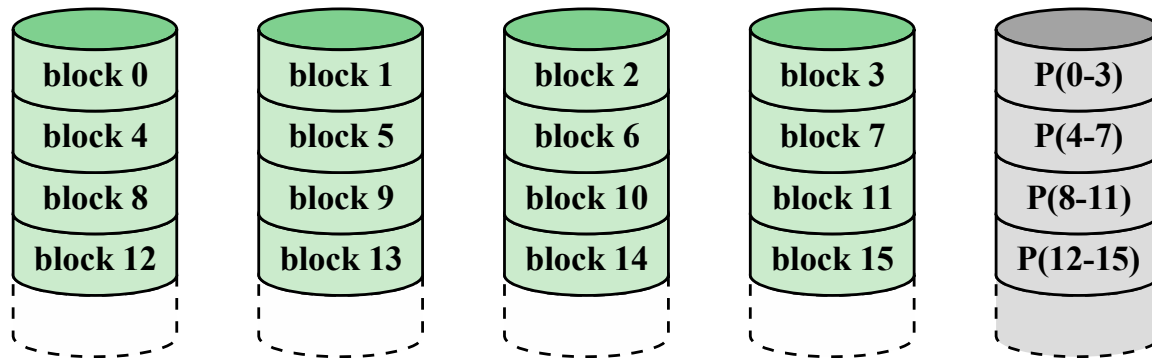


(c) RAID 2 (redundancy through Hamming code)

RAID Level 3, 4

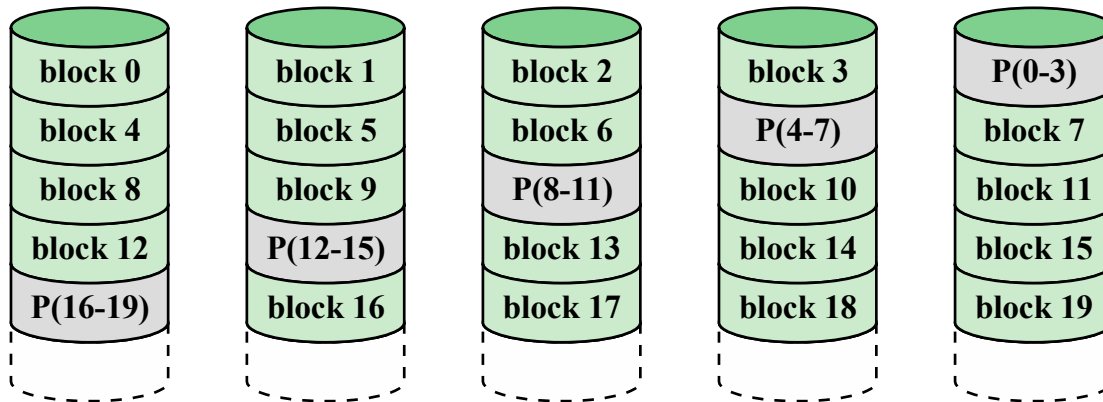


(d) RAID 3 (bit-interleaved parity)

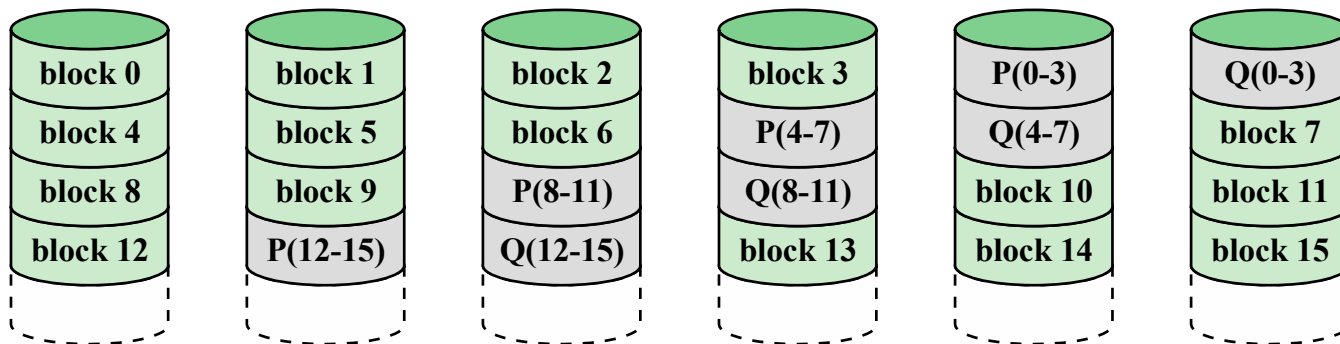


(e) RAID 4 (block-level parity)

RAID Level 5, 6

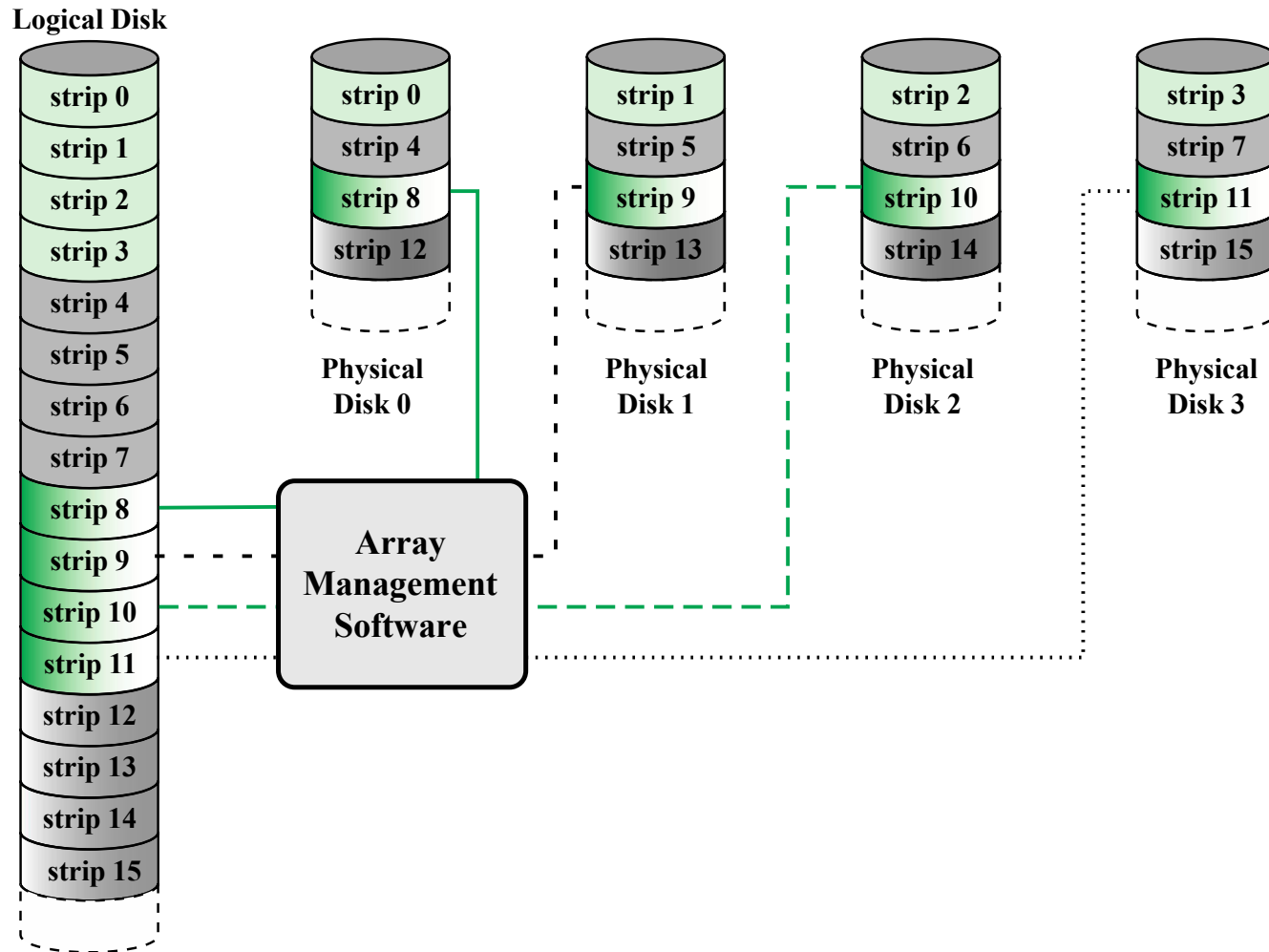


(f) RAID 5 (block-level distributed parity)



(g) RAID 6 (dual redundancy)

RAID Level 0 Data Mapping



RAID Comparison

Level	Advantages	Disadvantages	Applications
0	I/O performance is greatly improved by spreading the I/O load across many channels and drives No parity calculation overhead is involved Very simple design Easy to implement	The failure of just one drive will result in all data in an array being lost	Video production and Editing Image editing Pre-press applications Any application requiring high bandwidth
1	100% redundancy of data means no rebuild is necessary in case of a disk failure, just a copy to the replacement disk Under certain circumstances, RAID 1 can sustain multiple simultaneous drive failures Simplest RAID storage subsystem design	Highest disk overhead of all RAID types (100%) - inefficient	Accounting Payroll Financial Any application requiring very high availability
2	Extremely high data transfer rates possible The higher the data transfer rate required, the better the ratio of data disks to ECC disks Relatively simple controller design compared to RAID levels 3,4 & 5	Very high ratio of ECC disks to data disks with smaller word sizes - inefficient Entry level cost very high - requires very high transfer rate requirement to justify	No commercial implementations exist / not commercially viable

RAID Comparison

Level	Advantages	Disadvantages	Applications
3	Very high read data transfer rate Very high write data transfer rate Disk failure has an insignificant impact on throughput Low ratio of ECC (parity) disks to data disks means high efficiency	Transaction rate equal to that of a single disk drive at best (if spindles are synchronized) Controller design is fairly complex	Video production and live streaming Image editing Video editing Prepress applications Any application requiring high throughput
4	Very high Read data transaction rate Low ratio of ECC (parity) disks to data disks means high efficiency	Quite complex controller design Worst write transaction rate and Write aggregate transfer rate Difficult and inefficient data rebuild in the event of disk failure	No commercial implementations exist / not commercially viable
5	Highest Read data transaction rate Low ratio of ECC (parity) disks to data disks means high efficiency Good aggregate transfer rate	Most complex controller design Difficult to rebuild in the event of a disk failure (as compared to RAID level 1)	File and application servers Database servers Web, e-mail, and news servers Intranet servers Most versatile RAID level
6	Provides for an extremely high data fault tolerance and can sustain multiple simultaneous drive failures	More complex controller design Controller overhead to compute parity addresses is extremely high	Perfect solution for mission critical applications